

AI-Driven Data Research in Quantitative Finance

Corporate Project with WorldQuant Consulting (Beijing) Co., Ltd.

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Abstract

This project develops an end-to-end framework for converting SEC Item 1A “Risk Factors” disclosures into quantitative forecasting and portfolio-construction signals. Annual 10-K risk paragraphs are extracted from EDGAR filings, classified into a dynamically evolved macro-meso risk taxonomy, aggregated into firm-year risk exposure vectors, and used to construct semantic peer networks. The forecasting model is a topology-only Spatio-Temporal Graph Attention Network (ST-GAT): cosine similarity between risk exposure vectors determines whether a peer edge exists, but raw cosine weights are not fed directly into the graph attention layer. This produces a strict ablation between the full risk graph, $\mathbf{A}_t = \mathbf{I} + \mathbf{A}_{risk,t}$, and an identity baseline, $\mathbf{A}_t = \mathbf{I}$, which uses only each equity’s own financial features.

The forecasting panel covers 2006–2024, with chronological splitting into training years 2006–2016, validation years 2017–2020, and out-of-sample test years 2021–2024. Each firm-year node uses $F = 12$ backward-looking financial and market features, standardized using training-year statistics only, and predicts one-year-ahead July-to-June cumulative return and annualized volatility. The default dynamic-taxonomy Meso specification is mixed on the 2021–2024 OOS panel: ST-GAT slightly improves return MAE and return Spearman rank, but the identity baseline has lower averaged RMSE and stronger volatility rank correlation. A full-window sensitivity specification with a lower semantic-deviation threshold, $\theta = 0.40$, produces stronger Meso forecast evidence: averaged RMSE improves from 0.3523 to 0.3176, averaged MAE from 0.2546 to 0.2012, and volatility RMSE from 0.2082 to 0.1555 relative to its matched identity baseline. Paired Macro-level experiments are much denser but underperform identity baselines, showing that broad category coverage alone is not sufficient for useful information transfer.

Graph diagnostics show that taxonomy hyperparameters materially affect graph sparsity. In the 2021–2024 OOS period, the default Meso topology retains 32–96 off-diagonal directed edges per year, with active-node density below 0.16%. The $\theta = 0.40$ taxonomy retains

100–630 edges per year over the same period, with density from 0.17% to 1.20%. This indicates that the semantic-deviation threshold changes not only category counts but also the graph topology seen by the ST-GAT.

Finally, the project extends the forecasting model into an academic portfolio backtest. Annual ST-GAT predictions are held fixed over their July-to-June forecast window, while portfolios are rebalanced monthly using next-month execution timing, transaction costs of 10 bps, slippage of 5 bps, and SPY as benchmark. Under the default dynamic taxonomy, the ST-GAT return-only long-short portfolio achieves 7.31% annualized return, 8.34% annualized volatility, a Sharpe ratio of 0.88, Sortino ratio of 1.52, maximum drawdown of -8.06%, and beta of 0.30 against SPY. Its return-only top-minus-bottom spread is 16.26% annualized, with a conventional t -test $p = 0.080$ and HAC/Newey-West $p = 0.034$ before Benjamini-Hochberg adjustment. The $\theta = 0.40$ sensitivity variant improves forecast errors but does not uniformly improve portfolio ranking: its ST-GAT return-only spread is 9.33% annualized, below its identity-baseline spread of 11.68%. The defensible conclusion is therefore nuanced: textual risk topology can improve forecast and ranking diagnostics, but the portfolio alpha interpretation is sensitive to taxonomy evolution settings and short OOS windows.

Keywords – Text classification; Large language models; SEC Item 1A; Risk taxonomy; Graph Neural Networks; ST-GAT; Portfolio backtesting; Volatility forecasting; Risk similarity; Semantic peer networks; Volatility contagion.

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1 INTRODUCTION

1.1 Background and Motivation

Since 2005, the U.S. SEC has required public companies to disclose material risks in Item 1A “Risk Factors” of Form 10-K. Prior empirical studies show that these risk-factor disclosures contain market-relevant information and shape investors’ risk perceptions (Kravet and Muslu, 2013; Campbell et al., 2014; Hope et al., 2016). These narratives encode exposures to regulatory, operational, cyber, liquidity, and macroeconomic risks—information critical for valuation, portfolio construction, and systemic risk monitoring. However, the text is long, legalistic, and inconsistent across firms and years. Dyer et al. (2017) document that 10-K textual disclosure has expanded substantially over time and that boilerplate language has become increasingly common. Some firms substantially rewrite their risk profiles annually, while others exhibit exactly zero drift, copying boilerplate text. This unstructured format further hinders systematic analysis and motivates the need for structured NLP-based extraction.

1.2 Problem Statement

We address two core problems:

1. **Transformation:** How to reliably classify each risk paragraph into a consistent, interpretable taxonomy?
2. **Validation:** How to prove that the resulting quantitative signals have economic meaning, e.g., predict stock return or explain cross-sectional variation?

1.3 Research Questions

- **RQ1:** Can LLMs reliably classify risk disclosures into a hierarchical taxonomy?
- **RQ2:** Does a graph defined by textual risk similarity (ST-GAT) improve volatility and return prediction over an identity-matrix baseline that uses only each equity’s own data?
- **RQ3:** Do ST-GAT forecasts translate into economically meaningful dynamic portfolio performance under monthly rebalancing, transaction costs, and benchmark comparison?

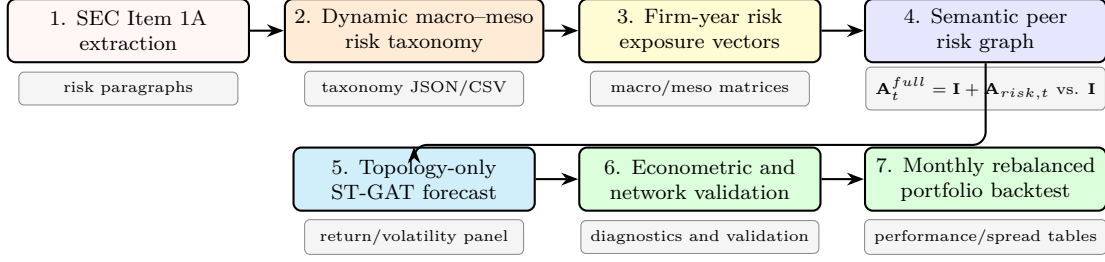


Figure 1: Conceptual end-to-end research pipeline. The diagram is schematic and summarizes the empirical sequence from disclosure text to taxonomy, graph construction, forecasting, and portfolio diagnostics.

1.4 Contributions

- **LLM-driven hierarchical taxonomy with dynamic evolution:** The project implements a macro-meso taxonomy evolution mechanism that can add new categories, merge emerging clusters into existing nodes, and reorganize parent macro assignments. Coverage diagnostics are interpreted as taxonomy-quality evidence rather than as standalone proof of forecasting value.
- **Topology-only ST-GAT with strict spatial ablation:** The main forecasting model uses $\mathbf{A}_t = \mathbf{I} + \mathbf{A}_{risk,t}$, where off-diagonal edges are selected by thresholded cosine similarity of risk exposure vectors. The identity baseline uses $\mathbf{A}_t = \mathbf{I}$, isolating the incremental contribution of textual peer topology.
- **Macro-Meso semantic-resolution and sensitivity comparison:** Macro-level graphs produce broad but overconnected peer topologies, while Meso-level graphs test whether finer semantic resolution improves information transfer. The full-window $\theta = 0.40$ Meso taxonomy produces a denser graph than the default Meso taxonomy and stronger OOS forecast-error reductions. This sensitivity result shows that taxonomy evolution settings are an empirical modeling choice, not a harmless preprocessing detail.
- **Dynamic portfolio extension:** Annual ST-GAT forecasts are converted into monthly rebalanced long-only and long-short portfolios. The default Meso ST-GAT return-only long-short portfolio achieves a Sharpe ratio of 0.88, maximum drawdown of -8.06%, and a 16.26% annualized top-minus-bottom spread with nominal but not multiple-testing-adjusted support.
- **Multi-model validation:** Integration of LLM taxonomy construction, ST-GAT forecasting, portfolio backtesting, and supplemental EGARCH-X/DAV, Fama-MacBeth, and network diagnostic modules. Source-table-limited econometric outputs are reported as methodology extensions rather than headline evidence.

1.5 Report Organization

The remainder of this report is organized as follows. Section 2 reviews financial textual analysis, SEC risk disclosure studies, graph neural networks, volatility modeling, and cross-sectional asset-pricing validation. Section 3 describes the data sources and preprocessing pipeline, including risk-factor extraction and the enhanced macro-financial feature matrix. Section 4 develops the dynamic risk-taxonomy methodology. Section 5 formally defines the ST-GAT architecture, including graph construction, identity-matrix ablation, spatial attention, temporal convolution, and masked multi-task loss. Section 6 specifies tensor construction, train-only standardization, active-mask logic, feature-target alignment, chronological splitting, and hyperparameter tuning. Section 7 reports forecasting results, convergence diagnostics, graph-density analysis, and taxonomy sensitivity. Section 8 introduces the dynamic portfolio backtesting extension, including monthly rebalancing, transaction costs, score ablations, and top-minus-bottom spread diagnostics. Sections 9 and 10 discuss limitations and conclude. Supplemental econometric baseline evaluations (DAV, Fama-MacBeth, and SAR models) and a representative sample of the evolved taxonomy are provided in the Appendices.

2 LITERATURE REVIEW

2.1 Textual Analysis in Finance: From Dictionaries to LLMs

Lexicon-based methods measure sentiment using pre-defined word lists but cannot capture context, negation, or industry-specific meanings (Loughran and McDonald, 2011; Li, 2010). Topic modeling methods such as Latent Dirichlet Allocation discover latent themes but require a pre-specified number of topics and ignore hierarchical structure (Blei et al., 2003; Bao and Datta, 2014). Transformer-based LLMs overcome many of these limitations by learning bidirectional contextual representations (Devlin et al., 2019; Liu et al., 2019; Beltagy et al., 2020). Recent financial-text benchmarks and working papers further motivate domain-specific LLM evaluation on SEC filings (Choi et al., 2026; Jiang et al., 2026).

Gap 2.1: No prior work systematically compares multiple LLM architectures (DeepSeek-R1, GPT-4o-Mini, Llama-3.1) for hierarchical risk taxonomy classification of Item 1A, nor evaluates multi-level prompting strategies.

2.2 Empirical Studies of SEC Item 1A Risk Factor Disclosures

Prior accounting and finance studies establish that SEC Item 1A risk-factor disclosures contain market-relevant information. Kravet and Muslu (2013) find that changes in risk disclosures are associated with investors' risk perceptions, while Hope et al. (2016) show that more specific risk disclosures trigger stronger market responses. Campbell et al. (2014) document the information content of mandatory risk-factor disclosures, and Dyer et al. (2017) document the rising length, boilerplate, and stickiness of 10-K text over time. Grundy and Petry (2021) apply sentence-level

LDA to S&P 1500 firms and show that quantified risk-factor text explains a meaningful fraction of cross-sectional risk variation. However, no prior study has systematically quantified year-over-year cosine drift in risk vectors or linked zero-drift boilerplate firms to the performance of graph-based predictive models.

Gap 2.2: The phenomenon of exactly zero drift (e.g., GE, NTAP) and its role in GNN underperformance has not been empirically established.

2.3 Graph Neural Networks (GNNs) in Financial Forecasting

Graph neural networks propagate information over non-Euclidean structures and have become a natural tool for modeling relational dependence (Veličković et al., 2018). Spatio-temporal graph attention architectures have also been applied to financial volatility forecasting and risk modeling by propagating information through graphs of interconnected assets (Kumar et al., 2024). Existing financial GNN studies typically construct graphs from industry classifications, historical correlations, supply-chain relations, or other static proxies. Niu et al. (2024) emphasize that static graph construction can fail to capture the evolving nature of financial relationships.

Gap 2.3: No prior study has constructed a graph directly from the semantic content of 10-K risk disclosures, systematically compared macro- versus meso-level graph resolutions, or performed a strict ablation study (replacing the graph with an identity matrix) to isolate the contribution of network topology.

2.4 Volatility Modeling with Textual Features

GARCH and EGARCH models capture volatility clustering and leverage effects in asset returns (Bollerslev, 1986; Nelson, 1991). Extensions such as GARCH-X and EGARCH-X allow exogenous variables to enter the conditional variance equation (Naimoli et al., 2023). A few studies have incorporated news-based sentiment or information volume, but the frequency mismatch between annual risk disclosures and daily volatility has received little attention.

Gap 2.4: No prior work systematically documents how annual risk scores, when repeated daily, are absorbed into the long-run variance intercept, nor provides a diagnostic framework (BIC, Diebold-Mariano, Mincer-Zarnowitz) to identify when textual features add value versus add noise.

2.5 Cross-Sectional Asset Pricing with Textual Features

The Fama-MacBeth two-step procedure is the standard methodology for testing whether a firm characteristic commands a cross-sectional risk premium (Fama and MacBeth, 1973). Gao (2016) proposes a text-implied risk factor using language-model embeddings, and later disclosure-risk factor studies extend traditional factor models with textual risk signals (Stevens Institute of Technology, 2020). Lopez-Lira (2019) also shows that textual risk factors can be informative for

the cross-section of returns. However, these approaches have not been applied to LLM-derived hierarchical risk scores from Item 1A.

Gap 2.5: The Fama-MacBeth framework has never been used to test whether LLM-based risk scores from 10-K filings explain cross-sectional variation in returns after controlling for size, momentum, and trailing volatility.

2.6 Research Gap and the Present Contribution

Synthesizing the existing literature, four interconnected gaps remain: (i) the absence of quantitative metrics for year-over-year textual drift linked to predictive performance; (ii) the lack of semantic-similarity graph architectures with rigorous spatial ablation protocols; (iii) the absence of a diagnostic framework to address the frequency mismatch between annual disclosures and daily volatility; and (iv) a lack of Fama-MacBeth validation for LLM-derived risk scores from SEC Item 1A.

The present research addresses these gaps by integrating an LLM-driven dynamic taxonomy, a multi-resolution topology-only ST-GAT architecture, and a suite of supplemental validation models. The empirical result is deliberately conservative. Under the default full-window taxonomy, the Meso ST-GAT improves return MAE and return Spearman rank but does not dominate the identity baseline on averaged OOS forecast metrics. Under the full-window $\theta = 0.40$ sensitivity taxonomy, the Meso graph model improves averaged RMSE from 0.3523 to 0.3176 and averaged MAE from 0.2546 to 0.2012 relative to its matched identity baseline, with the strongest gain in volatility RMSE. Macro-level graphs are included as paired sensitivity experiments and underperform identity baselines despite much higher graph density, reinforcing the need for semantic-resolution controls. The portfolio extension is economically suggestive but not conclusive: the default Meso ST-GAT return-only long-short strategy achieves a Sharpe ratio of 0.88 and a 16.26% annualized top-minus-bottom spread, while the $\theta = 0.40$ sensitivity variant has weaker portfolio ranking despite stronger forecast errors. Supplemental DAV peak diagnostics are retained as event-local evidence, while DAV BIC, Fama-MacBeth, and SAR modules are framed as validation extensions when archived source tables are unavailable. The central contribution is therefore the identity-matrix ablation and sensitivity workflow, not an unconditional claim that every textual graph specification produces stable alpha.

3 DATA & PRE-PROCESSING

3.1 Data Sources

The empirical design combines textual disclosures, market data, macro-market proxies, model outputs, and validation diagnostics. Table 1 summarizes the role of each dataset. The full experimental run underlying the report uses a 2006–2024 risk-year panel. Each dataset is documented by source, frequency, granularity, purpose, and transformation so that the research

design can be audited without confusing raw data, engineered features, model outputs, and validation diagnostics.

Table 1: Data architecture used by the final research pipeline.

Dataset	Source / Frequency	Granularity	Purpose and Transformations
SEC Item 1A risk text	SEC EDGAR 10-K filings; annual	Paragraph / firm-year	Extract Item 1A, remove table-of-contents noise, retain headings, subheadings, and paragraph text for taxonomy construction and classification.
Dynamic taxonomy outputs	Annual taxonomy construction outputs; 2006–2024	Risk factor / category / year	Builds a 2006 base taxonomy and evolves it through annual deviation detection, ADD/MERGE/REORGANIZE actions, and LLM-assisted labels.
Risk exposure matrices	Classified Item 1A paragraphs; annual	Firm-year $\times$ category	Converts paragraph labels into normalized Macro and Meso exposure vectors used for graph construction.
Equity market data	Yahoo Finance / <code>yfinance</code> ; daily	Ticker-day then firm-year	Builds backward-looking volatility, momentum, moving-average, liquidity, and forward target variables.
Macro-market proxies	Yahoo Finance proxies; daily	Year-level overlay	Adds S&P 500 return/volatility, VIX level/change, and 10-year Treasury yield proxy variables over the same feature window.
Graph snapshots	Exposure vectors and financial matrix; annual	Node-year graph	Constructs thresholded cosine-similarity peer graphs and identity baselines for ST-GAT evaluation.
Forecast outputs	ST-GAT forecasting procedure; annual panel	Firm-year prediction	Stores return and volatility predictions for ST-GAT and identity baseline models.
Portfolio data	Forecast panel and monthly prices; monthly	Strategy-month / holding	Converts annual forecasts into monthly re-balanced long-only and long-short academic backtests.
Validation outputs	Robust OOS, portfolio inference, DAV/FMB/SAR procedures	Table / figure	Supports OOS diagnostics, portfolio inference, and conservative appendix validation.

For clarity, each data source has a distinct econometric role. SEC Item 1A text is annual, paragraph-level, and firm-specific; it matters because it records management’s own description of risk exposure before the forecast target window. Equity prices and volumes are daily observations transformed into annual firm-year features and forward labels; they matter because they supply the quantitative target variables and the non-textual controls needed for a fair ablation. Macro-market proxies are daily market-wide series collapsed into the same July-to-June window; they matter because broad volatility, equity-market, and rate regimes affect all firms simultaneously. Graph snapshots are derived annual objects, not raw data; they matter because the ST-GAT tests whether firms with similar textual risk exposure should exchange information. Forecast

and portfolio outputs connect statistical prediction to economic usefulness under explicit timing and transaction-cost assumptions.

The temporal scope is also source-specific. The taxonomy outputs cover annual disclosures from 2006 through 2024. The ST-GAT tables use the 2006–2024 risk-year panel, with training years 2006–2016, validation years 2017–2020, and OOS years 2021–2024. Portfolio returns are monthly and are mapped from annual forecasts into their July-to-June holding windows. Supplemental DAV, Fama-MacBeth, and SAR modules are retained as validation designs unless their source tables are present and auditable.

3.2 Sample and Timing Convention

The project is an annual prediction exercise rather than an event-day study. Let t denote the risk disclosure year. The feature window is

$$\mathcal{D}_t^{feat} = \{\text{trading days from July 1 of } t \text{ through June 30 of } t + 1\}, \quad (1)$$

and the target window is

$$\mathcal{D}_t^{target} = \{\text{trading days from July 1 of } t + 1 \text{ through June 30 of } t + 2\}. \quad (2)$$

Thus, the model sees Item 1A risk topology and financial features available by the end of the feature window and predicts strictly forward return and volatility over the next July-to-June window. This convention aligns annual corporate disclosures with annual asset-pricing targets and prevents future target information from entering feature construction. Filing-date microstructure effects are outside the validated ST-GAT design.

3.3 SEC Item 1A Extraction Pipeline

The textual pipeline has four stages. First, the SEC EDGAR downloader identifies annual 10-K filings for the target CIK/ticker list and downloads the primary HTML document. Second, the parser isolates Item 1A by detecting “Item 1A Risk Factors” boundaries, removing table-of-contents references, and stopping at the next item boundary. Third, HTML formatting cues are used to separate main headings, subheadings, and paragraph bodies. The intuition is simple: a 10-K is not a flat text file; headings and subheadings carry important semantic context, so the classifier should see both the local paragraph and its surrounding risk heading. Fourth, the parser exports a paragraph-level master file with company, year, sequence number, heading fields, paragraph text, and a concatenated `full_text` field.

3.4 Financial Feature and Target Construction

Daily adjusted prices $P_{i,d}$ are converted into log returns

$$r_{i,d} = \log \left(\frac{P_{i,d}}{P_{i,d-1}} \right). \quad (3)$$

The feature-window realized volatility and downside volatility are

$$\sigma_{i,t}^{feat} = \sqrt{252} \text{ sd} \left(\{r_{i,d} : d \in \mathcal{D}_t^{feat}\} \right), \quad (4)$$

$$\sigma_{i,t}^{down} = \sqrt{252} \text{ sd} \left(\{r_{i,d} : d \in \mathcal{D}_t^{feat}, r_{i,d} < 0\} \right). \quad (5)$$

The moving-average feature compares the end-of-window price to the trailing 200-day average:

$$\text{MA200Ratio}_{i,t} = \frac{P_{i,\text{end}(t)}}{\frac{1}{200} \sum_{d \in \mathcal{D}_{t,200}^{feat}} P_{i,d}} - 1. \quad (6)$$

The eleven-month momentum variable is

$$\text{Momentum11M}_{i,t} = \frac{P_{i,\text{May } 31,t+1}}{P_{i,\text{July } 1,t}} - 1, \quad (7)$$

where the endpoint is chosen before the forward target window. Liquidity is measured by average dollar volume and its logarithmic transform:

$$\text{DollarVolume}_{i,t} = \frac{1}{|\mathcal{D}_t^{feat}|} \sum_{d \in \mathcal{D}_t^{feat}} P_{i,d} V_{i,d}, \quad \text{LogDollarVolume}_{i,t} = \log(1 + \text{DollarVolume}_{i,t}). \quad (8)$$

The two supervised targets are forward cumulative return and forward annualized realized volatility:

$$R_{i,t}^{target} = \frac{P_{i,\text{end}(\mathcal{D}_t^{target})}}{P_{i,\text{start}(\mathcal{D}_t^{target})}} - 1, \quad (9)$$

$$\sigma_{i,t}^{target} = \sqrt{252} \text{ sd} \left(\{r_{i,d} : d \in \mathcal{D}_t^{target}\} \right). \quad (10)$$

The market overlay uses the same formulas for S&P 500 return and volatility, while VIX and the 10-year Treasury yield proxy enter through feature-window averages and changes. These variables control broad market regime, volatility regime, and interest-rate environment so the graph model is not forced to infer all common macro variation from text alone.

3.5 Train-Only Imputation and Standardization

All preprocessing statistics are fitted using only the training years 2006–2016. For feature f , define the training median, mean, and standard deviation as $\tilde{x}_f^{train}$, μ_f^{train} , and σ_f^{train} . Missing

active-node features are imputed by

$$x_{i,t,f}^{imp} = \begin{cases} x_{i,t,f}, & x_{i,t,f} \text{ observed,} \\ \tilde{x}_f^{train}, & x_{i,t,f} \text{ missing,} \end{cases} \quad (11)$$

and standardized as

$$\tilde{x}_{i,t,f} = \frac{x_{i,t,f}^{imp} - \mu_f^{train}}{\max(\sigma_f^{train}, \epsilon)}. \quad (12)$$

Here ϵ is a small numerical guard used conceptually when a training feature has near-zero dispersion. The implementation replaces zero standard deviations with one. This design is central for research integrity: validation and OOS years never influence imputation or scaling.

4 DYNAMIC RISK TAXONOMY CONSTRUCTION

The taxonomy is the conceptual bridge between unstructured disclosure text and quantitative modelling. Its purpose is to convert heterogeneous Item 1A paragraphs into a stable, interpretable, and measurable hierarchy of corporate risk themes. A static taxonomy would make the downstream modelling problem easier to audit, but it would also impose the assumption that the language of corporate risk remains fixed from 2006 to 2024. This assumption is implausible in a period containing the global financial crisis, cloud-computing adoption, cybersecurity escalation, COVID-19, supply-chain disruptions, inflation shocks, cryptocurrency risk, and generative-AI disclosures. The methodology therefore uses a dynamic macro-meso taxonomy: it preserves continuity across years while allowing new disclosure language to modify the category system when the existing semantic structure no longer fits the data.

4.1 Motivation: Why a Dynamic Taxonomy is Needed

Fixed dictionaries are attractive because they are transparent, but they fail when corporate language changes. A category set designed before cloud computing, generative AI, COVID-19, or cryptocurrency risk may force genuinely new disclosures into broad residual labels. This phenomenon is semantic decay: the text remains observable, but the old category system no longer describes it precisely. In a forecasting problem, semantic decay is not merely a labelling inconvenience. If firms discussing materially different risks are assigned to the same broad category, their exposure vectors become artificially similar, the graph creates misleading peer links, and the ST-GAT attention mechanism aggregates information from the wrong neighborhood.

The hierarchy has two levels. **Macro** categories describe broad risk families, such as regulation, supply chain, financial reporting, or governance. **Meso** categories describe narrower themes within those families. The Meso layer is the primary forecasting input because it preserves enough semantic resolution to distinguish, for example, general regulatory risk from antitrust remedy risk, environmental compliance risk, concession-renewal risk, or sector-specific operating-

license risk. The Macro layer remains useful for interpretation and robustness checks, but the graph-forecasting problem requires a finer notion of textual peer similarity.

4.2 Text Encoding: From Risk Paragraphs to Vectors

Each risk paragraph p is first converted into a contextual string that preserves disclosure structure:

$$c_p = \text{concat} \left(\text{Heading}_p, \text{Subheading}_p, \text{RawText}_p \right). \quad (13)$$

The sentence-transformer encoder maps this text into a fixed-dimensional semantic vector:

$$\mathbf{e}_p = f_\phi(c_p) \in \mathbb{R}^{384}. \quad (14)$$

The encoder is **all-MiniLM-L6-v2**, a sentence-transformer model that maps each paragraph into a 384-dimensional embedding. It is smaller than maximum-capacity transformer encoders, but the trade-off is appropriate for a longitudinal taxonomy pipeline that must process hundreds of thousands of paragraphs. The important property is contextual semantic encoding: two paragraphs that discuss similar risks should point in similar directions in embedding space even if they do not use identical keywords. The pipeline compares those directions using cosine similarity:

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}. \quad (15)$$

4.3 Base-Year Construction

The base taxonomy is constructed from 2006 disclosures. The pipeline filters extremely long paragraphs, embeds the full-context strings, and applies a two-stage clustering design. The base year is not chosen because it is assumed to be economically special; it is chosen because it is the first full annual disclosure cross-section in the empirical panel. All later years then evolve this initial semantic map sequentially.

Macro clustering. The 384-dimensional embeddings are projected with Uniform Manifold Approximation and Projection (UMAP):

$$\mathbf{u}_p^{\text{macro}} = g_{\text{UMAP}}^{\text{macro}}(\mathbf{e}_p) \in \mathbb{R}^5. \quad (16)$$

HDBSCAN then assigns a broad cluster label,

$$z_p^{\text{macro}} = h_{\text{HDBSCAN}}^{\text{macro}}(\mathbf{u}_p^{\text{macro}}), \quad (17)$$

where $z = -1$ denotes noise. The default base-year hyperparameters are UMAP $n_neighbors = 15$, HDBSCAN macro $\text{min_cluster_size} = 50$, and $\text{min_samples} = 5$.

Meso clustering. Within each macro cluster m , the pipeline repeats dimensionality reduction

and density clustering:

$$\mathbf{u}_p^{meso} = g_{UMAP,m}^{meso}(\mathbf{e}_p) \in \mathbb{R}^8, \quad z_p^{meso} = h_{HDBSCAN,m}^{meso}(\mathbf{u}_p^{meso}). \quad (18)$$

The default Meso minimum cluster size is 15. This choice favors categories that have enough disclosure support to be meaningful rather than single-paragraph labels that would be unstable year to year. Methodologically, this is a bias–variance trade-off: smaller clusters preserve more specificity but increase the risk of fragile categories, while larger clusters improve stability but can blur economically different risks.

4.4 LLM-Assisted Labeling and Hybrid Centroids

For each discovered cluster k , a large language model (LLM) receives representative disclosure samples and returns a category name, a formal definition, and a short centroid summary. The centroid used for future classification combines empirical and semantic anchors:

$$\mathbf{c}_k = 0.7 \left(\frac{1}{|\mathcal{C}_k|} \sum_{p \in \mathcal{C}_k} \mathbf{e}_p \right) + 0.3 f_\phi(\text{Summary}_k). \quad (19)$$

The first term keeps the category grounded in actual disclosure text. The second term makes the centroid less sensitive to outlier paragraphs by anchoring it to a concise human-readable summary. This is a hybrid design: clustering discovers structure from data, while the LLM provides interpretable labels and semantic summaries.

Noise handling is deliberately conservative. If HDBSCAN does not assign a paragraph to a Meso cluster, the paragraph is assigned to the nearest Meso centroid:

$$k^*(p) = \arg \max_k \cos(\mathbf{e}_p, \mathbf{c}_k). \quad (20)$$

This produces complete exposure vectors for downstream modeling while preserving the ability to audit which points were originally density-clustering noise.

4.5 Year-over-Year Dynamic Evolution

For every year after 2006, the previous year’s taxonomy is loaded and the new year’s disclosures are embedded. Each new paragraph is first classified by nearest Meso centroid:

$$k_{p,t}^* = \arg \max_{k \in \mathcal{K}_{t-1}} \cos(\mathbf{e}_{p,t}, \mathbf{c}_{k,t-1}), \quad s_{p,t}^* = \max_{k \in \mathcal{K}_{t-1}} \cos(\mathbf{e}_{p,t}, \mathbf{c}_{k,t-1}). \quad (21)$$

A paragraph is marked as a semantic deviation when

$$D_{p,t} = \mathbf{1}\{s_{p,t}^* < \theta\}, \quad \theta = 0.45. \quad (22)$$

Here $\mathbf{1}\{\cdot\}$ is an indicator function. The default threshold is $\theta = 0.45$ and is recorded in the annual run logs.

When enough deviations exist, the deviation embeddings are clustered with UMAP and HDB-SCAN. For each deviation cluster q , the LLM receives representative texts, the top three nearest existing categories, and a taxonomy summary. It then selects one of three implemented actions:

- **ADD:** create a new Meso category under an appropriate Macro parent.
- **MERGE:** absorb the cluster into an existing Meso category and revise its centroid.
- **REORGANIZE:** move an existing Meso category to a different Macro parent.

A fourth governance action, **REMOVE**, is conceptually useful for stale categories but is not used in the reported empirical pipeline. Keeping this distinction explicit avoids overstating the taxonomy engine: the current design can add, merge, and reorganize categories, while deletion remains a conservative future extension.

The ADD and MERGE centroid revisions are:

$$\mathbf{c}_{new}^{ADD} = 0.7 \left(\frac{1}{|\mathcal{D}_q|} \sum_{p \in \mathcal{D}_q} \mathbf{e}_{p,t} \right) + 0.3 f_\phi(\text{Name}_{new}), \quad (23)$$

$$\mathbf{c}_{k,t} = 0.8 \mathbf{c}_{k,t-1} + 0.2 \left(\frac{1}{|\mathcal{D}_q|} \sum_{p \in \mathcal{D}_q} \mathbf{e}_{p,t} \right). \quad (24)$$

The MERGE rule is intentionally slow-moving: it lets existing categories absorb moderate semantic drift without letting one noisy year overwrite the historical meaning of the category.

4.6 Risk Exposure Vectors

After classification, each firm-year is represented by normalized paragraph shares. Let $n_{i,t,k}$ be the number of Item 1A paragraphs for firm i in year t assigned to category k , and let $N_{i,t} = \sum_k n_{i,t,k}$. The exposure vector is

$$r_{i,t,k} = \frac{n_{i,t,k}}{N_{i,t}}, \quad \mathbf{r}_{i,t} = (r_{i,t,1}, \dots, r_{i,t,K}). \quad (25)$$

Macro vectors aggregate by broad parent category; Meso vectors retain the granular category level. These vectors are the bridge between textual NLP and quantitative finance: they convert qualitative disclosure emphasis into numerical exposures that can define graph edges, regressions, and portfolio signals.

4.7 Empirical Taxonomy Evolution Diagnostics

The default dynamic taxonomy is supported by annual run logs from 2006 through 2024. The base 2006 run contains 3,382 risk factors, 17 Macro clusters, 126 Meso clusters, 77.23% macro non-noise coverage, and 100% final Meso assignment after nearest-centroid noise resolution. The evolved taxonomy reaches 475 Meso categories by 2024. Across the annual logs, 349 ADD actions and 56 MERGE actions are recorded, with no REORGANIZE actions executed.

Table 2: Selected diagnostics from the default dynamic-taxonomy annual run logs.

Year	Deviation Rate	ADD	MERGE	Final Meso Categories
2006 base	—	—	—	126
2008	18.70%	64	2	190
2012	4.68%	25	10	243
2016	3.60%	61	7	365
2020	1.32%	10	1	459
2024	0.86%	3	2	475

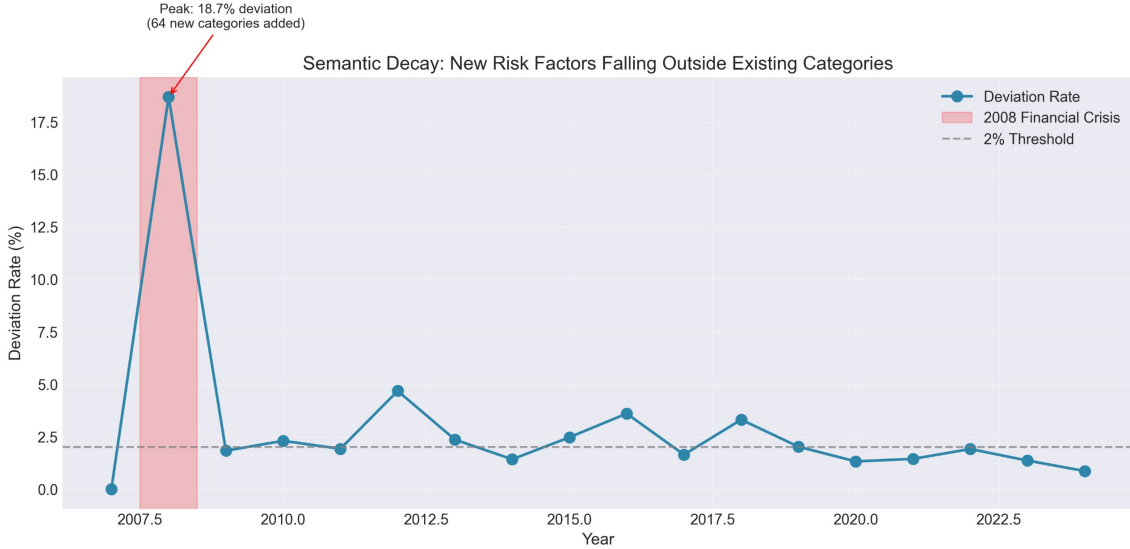


Figure 2: Semantic-deviation rate in the default dynamic taxonomy. The 2008 financial crisis generates the largest deviation spike, consistent with a regime in which new risk language falls outside pre-existing categories.

The most important interpretation is not that every new category is economically priced; it is that annual disclosures contain detectable semantic novelty. The 2008 spike is a useful stress test because crisis-related disclosures introduced language that the 2006–2007 taxonomy could not fully absorb.

4.8 Full-Window Sensitivity Design

A base-year-only sensitivity table can indicate whether the initial clustering solution is stable, but it cannot answer the stronger research question in this project: whether taxonomy hyperparameters remain consequential after annual evolution from 2006 through 2024, and whether downstream ST-GAT and portfolio conclusions are robust to the resulting alternative exposure vectors. The sensitivity analysis is therefore defined over complete taxonomy paths, not isolated base-year clusterings.

The implemented sensitivity grid now contains the default taxonomy plus expanded Macro, Meso, UMAP, HDBSCAN, centroid-weight, merge-weight, LLM-sampling, and random-seed variants. In compact notation, each variant is indexed by

$$h = (m_M, m_m, s_M, s_m, n_M, n_m, d_M, d_m, \theta, w_{LLM}, \rho, q, \xi), \quad (26)$$

where m denotes HDBSCAN minimum cluster size, s denotes HDBSCAN minimum samples, n denotes UMAP neighbor count, d denotes UMAP dimensionality, θ denotes the semantic-deviation threshold, w_{LLM} denotes the LLM-summary weight in centroid construction, ρ denotes the old-centroid retention weight in MERGE centroid revision, q denotes the number of representative paragraphs sent to the LLM prompt, and ξ denotes the random seed. The default setting is

$$h_0 = (50, 15, 5, 3, 15, 8, 5, 8, 0.45, 0.30, 0.80, 5, 42). \quad (27)$$

Table 3: Expanded full-window taxonomy sensitivity axes implemented in the pipeline.

Axis	Tested values or named variants	Role
Macro HDBSCAN cluster size	20, 30, 40, 50, 70, 90	Base-year broadness
Meso HDBSCAN cluster size	8, 10, 15, 20, 25	Within-macro granularity
Base UMAP neighbor count	8, 10, 15, 20, 30	Local/global manifold balance
Deviation threshold θ	0.35, 0.40, 0.45, 0.50, 0.55	Novelty gate
HDBSCAN minimum samples	Macro 3/5/8; Meso 1/3/5	Density conservatism
UMAP dimensionality and Meso neighbors	3, 5, 8, 10 components; Meso neighbors 5/8/12	Projection geometry
Centroid and MERGE weights	$w_{LLM} = 0.20, 0.30, 0.40$; $\rho = 0.75, 0.80, 0.90$	Semantic anchoring and drift speed
LLM prompt sample size and seed	5-sample default; seed 42/43	Label stability

For a variant $h \in \mathcal{H}$, the intended full-window taxonomy path is

$$\mathcal{K}_{2006}^h \rightarrow \mathcal{K}_{2007}^h \rightarrow \cdots \rightarrow \mathcal{K}_{2024}^h, \quad (28)$$

where each year loads the previous year’s evolved taxonomy rather than restarting from the base year. Each completed path produces paired Macro and Meso exposure matrices. Those matrices are then passed into the same ST-GAT evaluation design, so the downstream robustness estimands are:

$$\begin{aligned} \Delta_{error}^{h,\ell} &= \text{Error}(\text{Identity}; \mathcal{K}^{h,\ell}) - \text{Error}(\text{ST-GAT}; \mathcal{K}^{h,\ell}), \\ \Delta_{rank}^{h,\ell} &= \rho(\text{ST-GAT}; \mathcal{K}^{h,\ell}) - \rho(\text{Identity}; \mathcal{K}^{h,\ell}). \end{aligned} \quad (29)$$

where $\ell \in \{\text{Macro}, \text{Meso}\}$. Positive error improvement means ST-GAT has lower RMSE or MAE than the matched identity baseline; positive rank improvement means ST-GAT has higher Spearman rank correlation. In plain English, the sensitivity test asks whether the graph advantage survives when the taxonomy-building parameters and semantic resolution are changed. Among the full-window variants with complete taxonomy, paired Macro/Meso forecast, and portfolio evidence in this manuscript, the $\theta = 0.40$ path is the central sensitivity comparison. It processes the same 578,516 risk factors as the default $\theta = 0.45$ path but admits fewer semantic deviations. It produces 258 final Meso categories by 2024, compared with 475 under the default taxonomy. The lower threshold is stricter: a paragraph must be farther from existing centroids before it can trigger deviation clustering.

Table 4: Full-window taxonomy sensitivity: default dynamic taxonomy versus $\theta = 0.40$ variant.

Variant	Final Meso	ADD	MERGE	REORGANIZE
Default $\theta = 0.45$	475	349	56	0
Sensitivity $\theta = 0.40$	258	132	69	3

Figure 3 shows why this parameter matters. The default threshold accepts more year-over-year novelty and therefore accumulates more categories. The $\theta = 0.40$ variant absorbs more new paragraphs into existing semantic centroids, producing a coarser but less fragmented Meso space. This is not simply a question of interpretability. Because graph edges are constructed from cosine similarity between exposure vectors, a more fragmented taxonomy can reduce overlap between firms and create an extremely sparse peer graph. A moderately coarser taxonomy can increase graph connectivity and improve information sharing, especially for volatility forecasting.

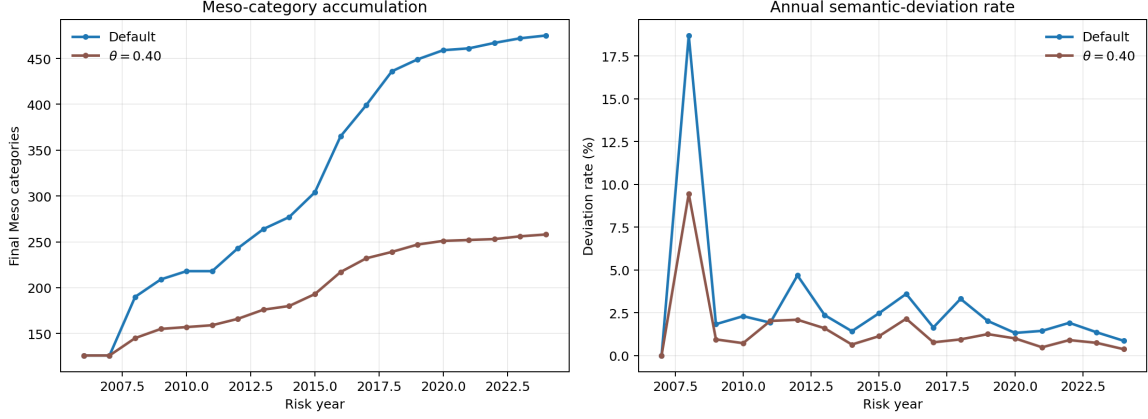


Figure 3: Full-window comparison between the default dynamic taxonomy and the $\theta = 0.40$ sensitivity path. The lower threshold produces fewer final Meso categories and lower annual semantic-deviation rates because new paragraphs must be more distant from existing centroids before triggering taxonomy evolution.

4.9 Best-Tested Taxonomy Coverage and Topological Structure

Among the completed downstream paths, the $\theta = 0.40$ taxonomy is the best forecasting taxonomy because it is the only tested path where the Meso ST-GAT improves averaged RMSE, averaged MAE, and averaged Spearman rank relative to its matched identity baseline. Its interpretability is different from the default taxonomy. The default path is more granular: by 2024, its largest Macro family is Regulatory and Operational License Risk, covering 29.57% of assigned paragraphs, followed by Property Operations & Revenue Dynamics at 22.03% and Corporate Governance & Anti-Takeover Provisions at 21.91%. The $\theta = 0.40$ path is more concentrated: Operational and External Event Risk covers 56.21% of 2024 paragraphs, followed by Corporate Governance & Anti-Takeover Provisions at 8.80%, Market and Credit Risk Management at 6.30%, and Accounting Changes and Share-Based Compensation at 5.55%.

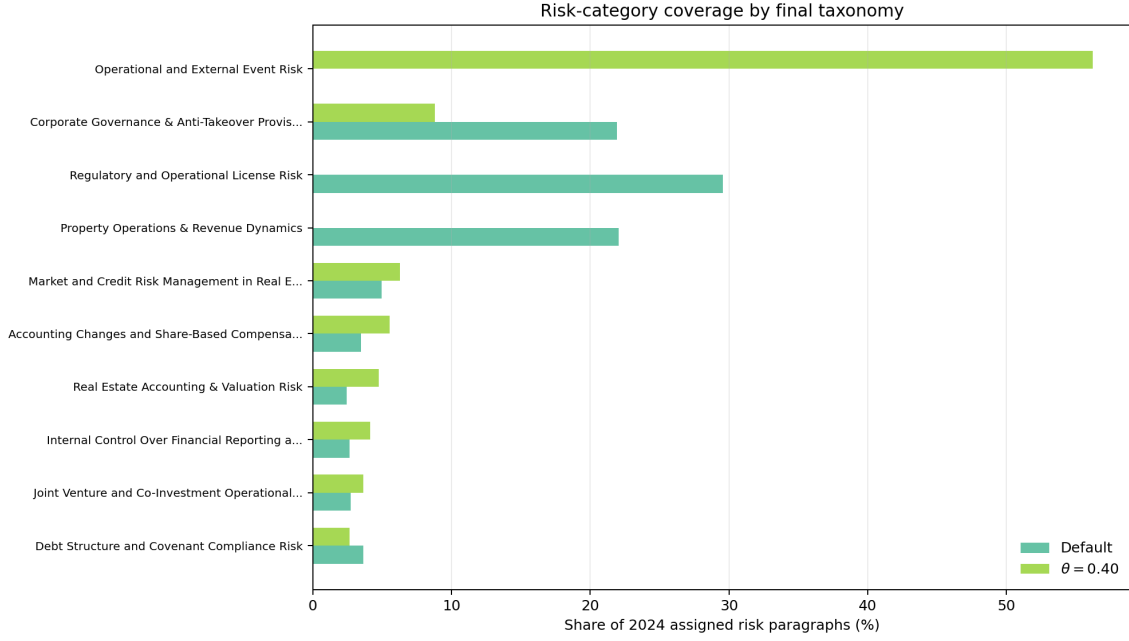


Figure 4: Coverage comparison for the final 2024 default and $\theta = 0.40$ taxonomies. Coverage is measured as the share of assigned Item 1A risk paragraphs in each Macro category.

This coverage shift explains why the two taxonomies behave differently. The default taxonomy spreads risk text across many narrower categories, preserving detailed interpretation but reducing overlap between firm exposure vectors. The $\theta = 0.40$ taxonomy covers more text with fewer broad Macro families and fewer Meso leaves. This coarser coverage makes the exposure vectors smoother and raises graph connectivity, but it also increases the risk of grouping economically distinct narratives under a common parent. In other words, the best-tested taxonomy is not simply “more accurate” in a linguistic sense; it chooses a different point on the coverage–resolution frontier.

The topology view in Figure 5 makes this frontier visible. Macro nodes are sized by 2024 paragraph coverage and Meso nodes show the largest child themes under the dominant Macro families. The graph is not a forecasting graph between firms; it is the taxonomy graph itself. It motivates the firm-graph analysis in Section 5 because it shows how a small change in semantic-deviation governance can reshape the hierarchy that later determines firm-to-firm similarity.

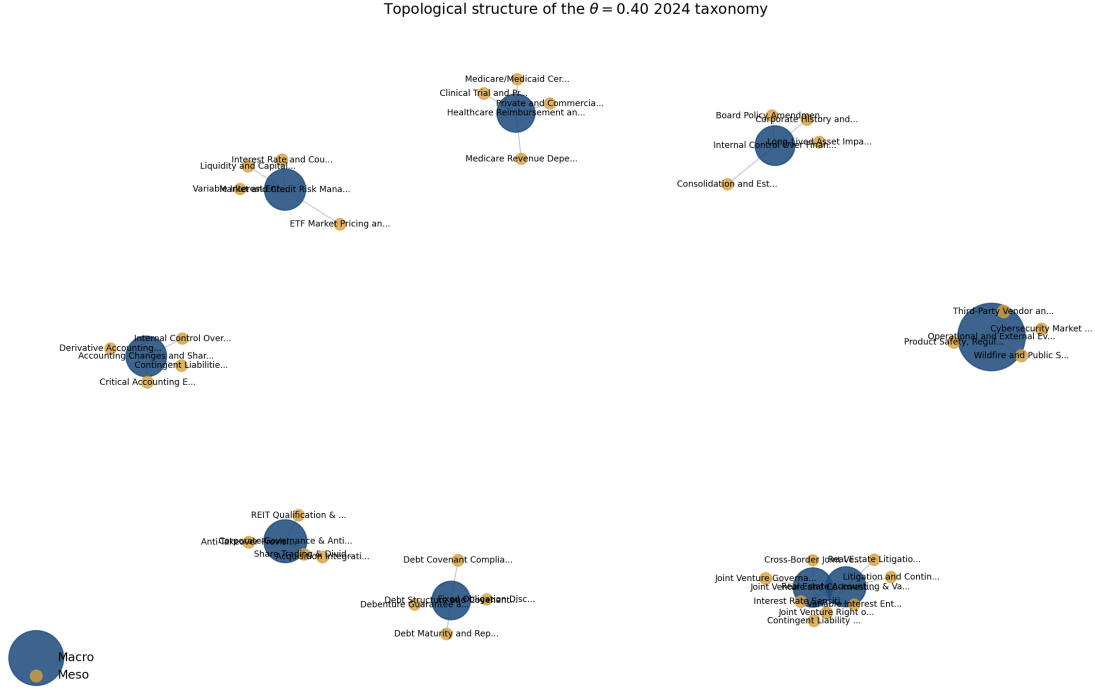


Figure 5: Topological structure of the best-tested $\theta = 0.40$ 2024 taxonomy. Macro nodes are broad risk families and Meso nodes are the largest child categories within the most-covered families.

The full sensitivity grid is therefore treated as a downstream experimental design rather than a cosmetic NLP ablation. For every completed taxonomy path h , the empirical workflow records category growth, deviation rate, paired Macro/Meso graph density, forecast-error improvement relative to the identity baseline, rank-correlation improvement, long-short Sharpe ratio, and top-minus-bottom spread. This makes taxonomy selection a disciplined model-selection problem: a taxonomy is preferable only if it improves both semantic coherence and downstream economic validation without inducing excessive category fragmentation or overconnected broad-category graphs.

4.10 Firm-Level Dynamic Taxonomy Validation: Las Vegas Sands

The firm-level case study is an exploratory data analysis rather than a forecast result. Its purpose is to ask whether the LLM-assisted taxonomy records economically interpretable movements in a firm’s disclosure vector. This is a face-validity check for the classification layer: if changes in a firm’s risk-vector trajectory coincide with identifiable business events, then the semantic graph used in Section 5 has a plausible economic object to learn from. If the vectors were pure taxonomy noise, the ST-GAT graph would be only a statistical construction.

Las Vegas Sands (LVS) is a useful audit case because its disclosure history contains large, externally observable shocks: the Marina Bay Sands expansion agreement in 2019, the COVID-19 disruption to Macao, Singapore, and Las Vegas operations in 2020, the agreement to sell its Las Vegas properties in 2021, and the Macao gaming concession renewal process completed in December 2022 (Las Vegas Sands, 2019, 2021a,b; Macao SAR Government, 2022; Sands

China, 2022). Figure 6 tracks the $\theta = 0.40$ Macro and Meso exposure paths and adds vector diagnostics. The main pattern is economically coherent. LVS moves away from early financial-reporting and debt-disclosure categories toward operational, gaming-competition, concession, pandemic, and brand/reputation categories as its risk profile becomes more concentrated in Asian integrated-resort operations.

Table 5: Event-aligned taxonomy validation for LVS. Exposure shares are computed from the reconstructed $\theta = 0.40$ Meso vectors.

Year	External business context	Taxonomy movement	Interpretation
2019	LVS announced a Marina Bay Sands expansion agreement and its 2019 filing entered the COVID-19 news window shortly before the February 2020 Form 10-K filing.	The vector distance from the prior available filing is 0.506. Pandemic/business-interruption exposure reaches 7.5%, while debt maturity is the largest single Meso category at 11.8%.	The classifier separates the growth-investment and travel-disruption narratives instead of placing all changes into a generic operational bucket.
2022	The Las Vegas property sale had been announced in March 2021 and the Macao concession process culminated in a new 10-year contract in December 2022.	The vector distance is 0.433. Casino resort competition is 18.2%, pandemic interruption is 12.1%, and Macao gaming tax plus debt-covenant categories each appear at 6.1%.	The taxonomy records the transition from a diversified Las Vegas/Macao/Singapore operator toward an Asia-concession and recovery-risk profile.
2023– 2024	The post-concession period begins and pandemic restrictions fade from the headline operating environment.	Vector distances fall to 0.197 and 0.105. Casino resort competition remains the leading Meso category at 14.7% and 24.2%, while explicit pandemic exposure is no longer a top category in 2024.	The vector stabilizes after the concession reset, consistent with a new operating regime rather than a continuing one-off shock.

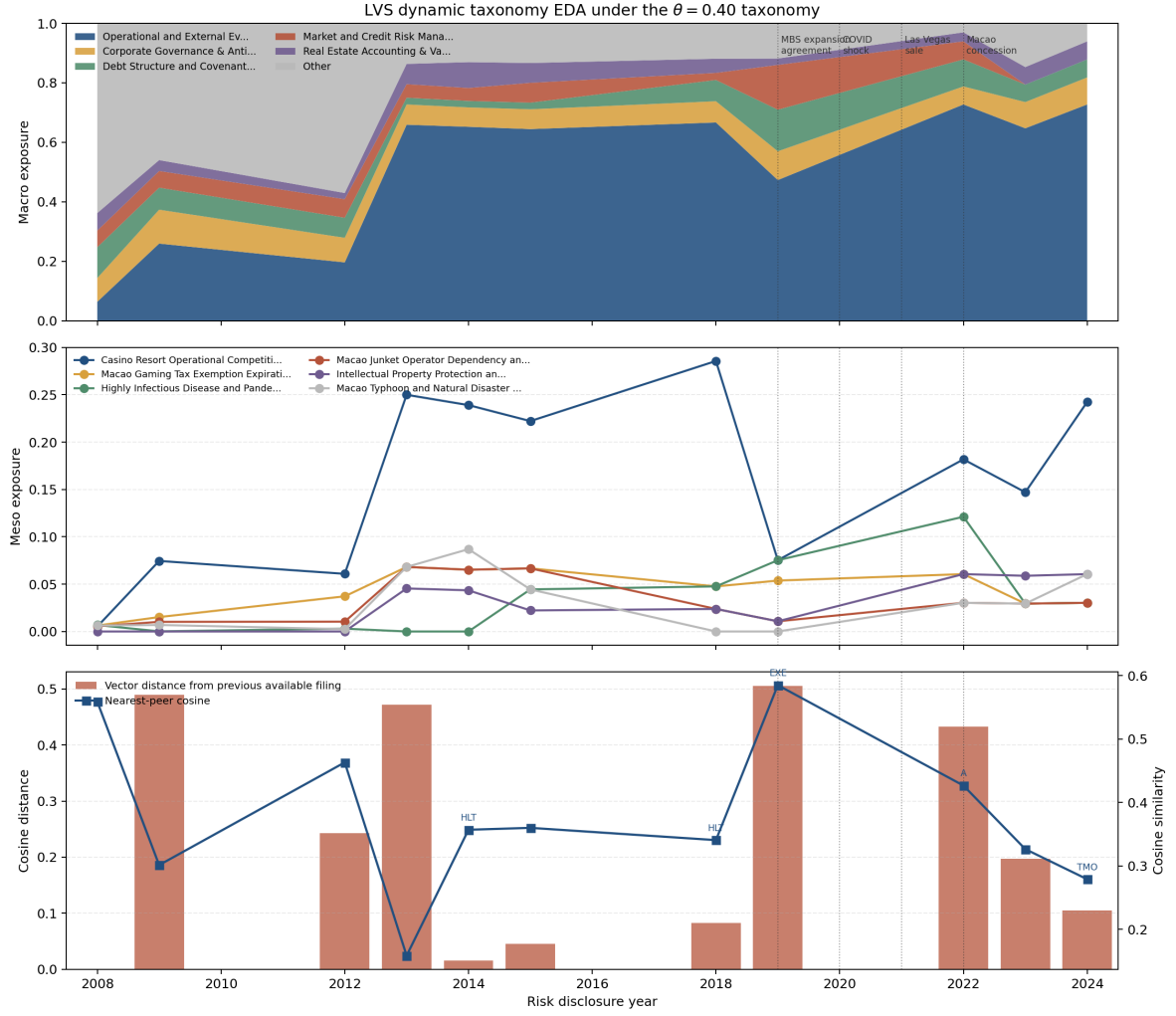


Figure 6: LVS dynamic taxonomy EDA under the best-tested $\theta = 0.40$ taxonomy. The upper panel shows Macro exposure shares, the middle panel shows selected Meso trajectories, and the lower panel shows year-to-year Meso-vector movement together with the nearest semantic-peer similarity.

The network view in Figure 7 provides the bridge to the ST-GAT design. Each point is a firm-year Meso exposure vector; the red path is LVS; blue squares are the nearest semantic peers in selected years. The nearest neighbors are not simply a static industry screen. In 2014–2018, Hotel and lodging exposure makes HLT the closest peer. In the 2019 and 2022 shock windows, the nearest peers change as the LVS vector moves toward other firms with overlapping financing, operational-disruption, development, and regulatory language. This is exactly the kind of time-varying peer relation the graph model is meant to test. The case study therefore motivates, but does not prove, the Section 5 graph construction: predictive value still has to come from the identity-matrix ablation in Section 7.

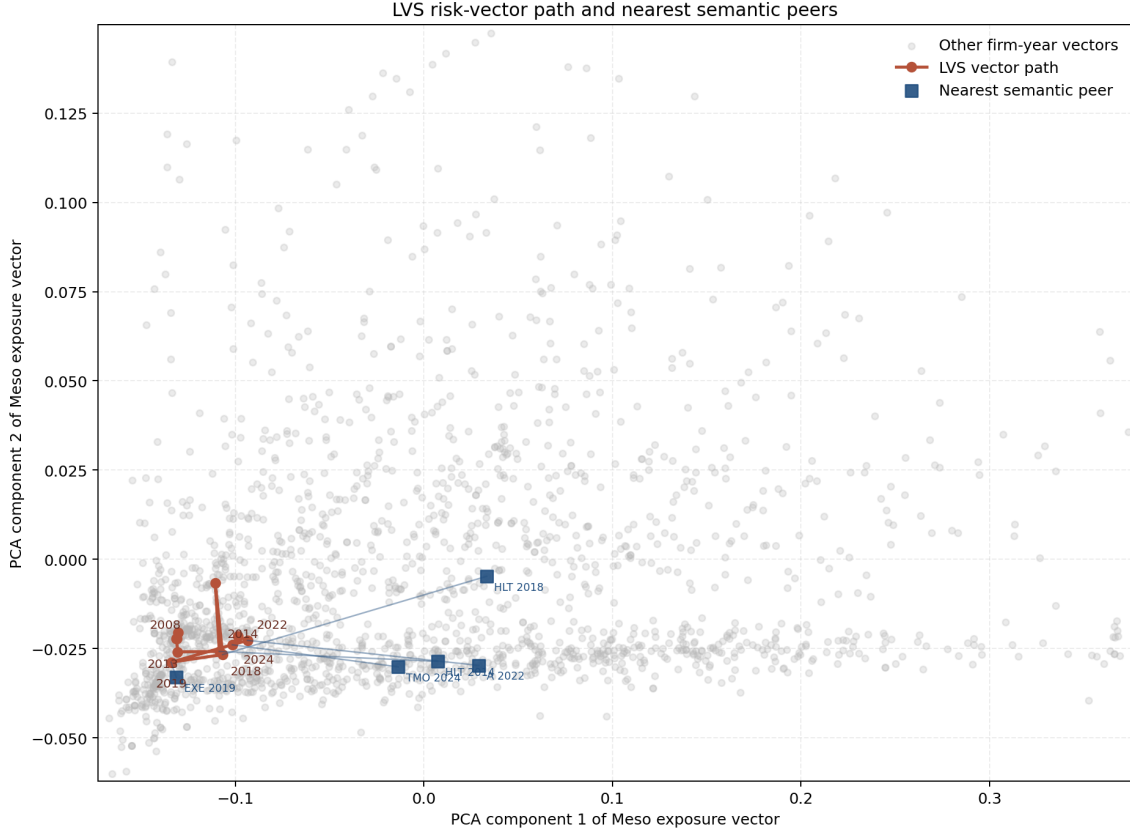


Figure 7: LVS risk-vector path and selected nearest semantic peers in the Meso exposure-vector space. The two displayed axes are PCA coordinates used only for EDA visualization; ST-GAT edges are built from cosine similarity in the full taxonomy-vector space.

This single-firm analysis does not establish classification accuracy in the supervised-learning sense, because there are no hand-labelled ground-truth risk categories. It does provide an audit trail for face validity: the LLM-assisted labels, the normalized exposure vector, and the nearest-neighbor graph all move in directions that can be interpreted against external business events. That is the empirical reason for carrying the taxonomy vectors forward into the ST-GAT instead of stopping at a static disclosure summary.

4.11 Validation Limits

Several taxonomy limitations remain. The default deviation threshold $\theta = 0.45$ is only one governance choice; the $\theta = 0.40$ sensitivity path shows that category counts, graph density, forecast errors, and portfolio ranking can all change when the threshold changes. LLM labels can introduce semantic style drift, even when centroids remain data-driven. Category deletion is not implemented, so the taxonomy may accumulate stale categories over long horizons. Finally, firm-level case studies validate interpretability but should not be treated as cross-sectional proof. These limitations are carried into the empirical design by requiring identity-baseline ablations, graph-density diagnostics, and downstream taxonomy sensitivity checks.

5 MATHEMATICAL FORMULATION OF THE SPATIO-TEMPORAL ARCHITECTURE

The predictive objective of the ST-GAT model is to determine whether semantic similarity in annual SEC Item 1A risk disclosures provides cross-sectional information beyond each firm’s own financial trajectory. The architecture is therefore designed as a strict spatial ablation. The full model uses a dynamic risk-similarity topology, while the baseline uses only self-loops. This section defines the graph, attention mechanism, temporal convolution layer, masking logic, and multi-task loss used for one-year-ahead return and volatility prediction.

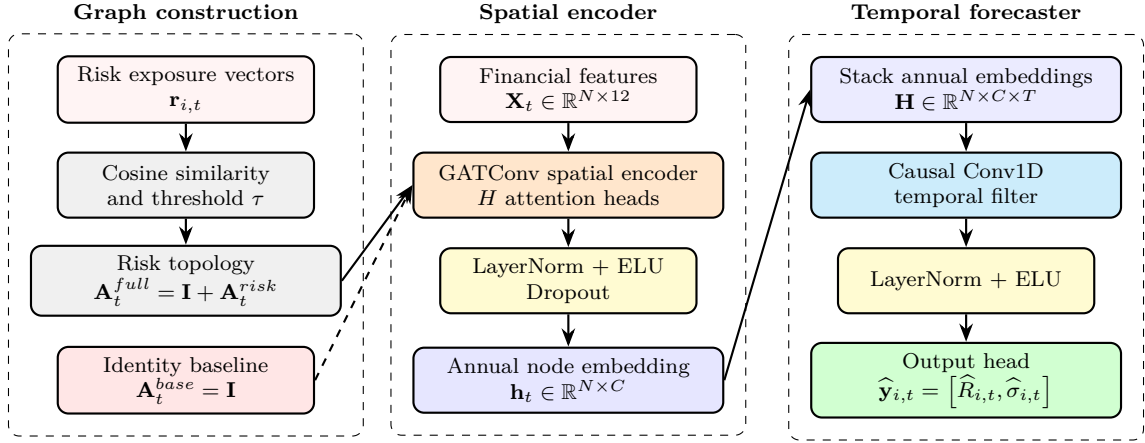


Figure 8: ST-GAT block architecture and identity-matrix ablation. Risk exposure vectors define the sparse graph topology, financial node features enter the spatial GAT encoder, and annual embeddings are passed through a causal temporal forecaster to predict one-year-ahead return and volatility. The full model uses textual peer topology, whereas the identity baseline removes all off-diagonal links. Cosine similarity determines edge existence, while attention weights are learned by GATConv.

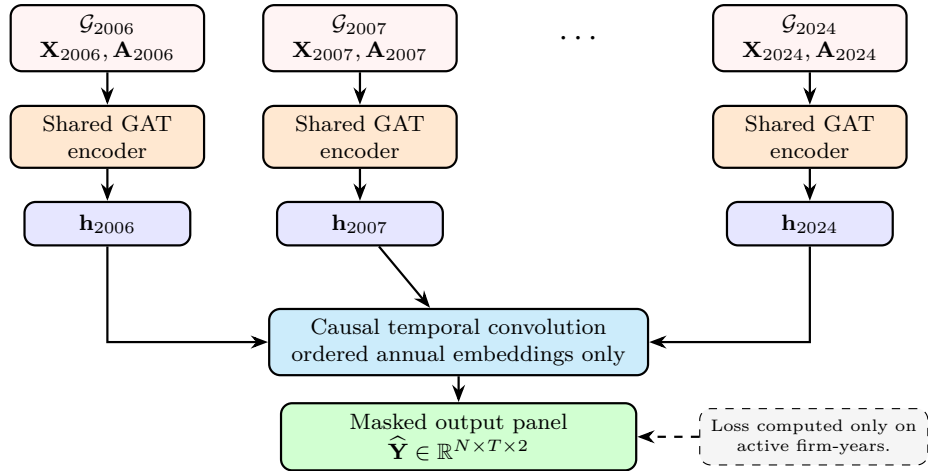


Figure 9: Temporal processing of annual graph snapshots. The shared GAT encoder is applied to each annual graph, and a causal temporal convolution produces masked return-volatility predictions across the full panel.

5.1 Dynamic Risk-Similarity Graph Definition

For each year t , the financial universe is represented as an attributed graph:

$$\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathbf{A}_t, \mathbf{X}_t), \quad (30)$$

where $\mathcal{V} = \{v_1, \dots, v_N\}$ is the global node universe, $\mathcal{E}_t$ is the annual edge set, $\mathbf{A}_t \in \mathbb{R}^{N \times N}$ is the adjacency matrix, and $\mathbf{X}_t \in \mathbb{R}^{N \times F}$ is the node feature matrix. In the final implementation, $F = 12$ financial and macro-market features are used for each firm-year.

For each firm i and year t , the risk exposure vector $\mathbf{r}_{i,t}$ is constructed from normalized paragraph shares:

$$r_{i,k,t} = \frac{\text{number of Item 1A paragraphs of firm } i \text{ assigned to risk category } k \text{ in year } t}{\text{total Item 1A paragraphs of firm } i \text{ in year } t}. \quad (31)$$

Macro graphs use first-level risk categories, while Meso graphs use second-level risk categories. The off-diagonal semantic similarity between firms i and j is computed using cosine similarity:

$$s_{ij,t} = \frac{\mathbf{r}_{i,t}^\top \mathbf{r}_{j,t}}{\|\mathbf{r}_{i,t}\|_2 \|\mathbf{r}_{j,t}\|_2}. \quad (32)$$

A risk edge is retained only if similarity exceeds an optimizable threshold τ :

$$A_{ij,t}^{risk} = \begin{cases} 1, & i \neq j \text{ and } s_{ij,t} \geq \tau, \\ 0, & \text{otherwise.} \end{cases} \quad (33)$$

The main reported specification is topology-only: cosine similarity determines whether an edge exists, but raw cosine weights are not directly passed into the GAT attention mechanism. This design was selected after weighted-edge attention was found to be less stable empirically. Edge weights are retained for diagnostics and optional ablations, but the primary model allows the GAT layer to learn propagation weights from node features and graph topology.

5.2 Full Graph versus Identity Baseline

The final spatial ablation is defined as:

$$\mathbf{A}_t^{full} = \mathbf{I}_N + \mathbf{A}_t^{risk}, \quad (34)$$

while the baseline is:

$$\mathbf{A}_t^{base} = \mathbf{I}_N. \quad (35)$$

Therefore, the full ST-GAT receives each firm’s own information and information from textually similar peers, while the baseline receives only each firm’s own information. This establishes a clean test of whether the NLP-derived risk topology contributes predictive information beyond isolated firm-level features.

Table 6: Adjacency structures used in the ST-GAT ablation.

Model	Adjacency	Interpretation
Full ST-GAT	$\mathbf{A}_t^{full} = \mathbf{I}_N + \mathbf{A}_t^{risk}$	Own-firm information plus peer spillovers from firms with similar 10-K risk exposure vectors.
Identity Baseline	$\mathbf{A}_t^{base} = \mathbf{I}_N$	Own-firm information only; all off-diagonal textual peer links are removed.

5.3 Spatial Graph Attention Mechanism

For each annual graph snapshot, a graph attention layer transforms node features into spatial embeddings, following the attention-based neighborhood aggregation principle of Graph Attention Networks (Veličković et al., 2018).

$$e_{ij,t} = \text{LeakyReLU} \left(\mathbf{a}^\top [\mathbf{W}\mathbf{x}_{i,t} \parallel \mathbf{W}\mathbf{x}_{j,t}] \right), \quad (36)$$

where $\mathbf{W}$ is a learnable projection, $\mathbf{a}$ is the attentional vector, and $\parallel$ denotes concatenation. The normalized attention weight is:

$$\alpha_{ij,t} = \frac{\exp(e_{ij,t})}{\sum_{k \in \mathcal{N}_t(i)} \exp(e_{ik,t})}. \quad (37)$$

The spatial embedding for node i is then:

$$\mathbf{h}_{i,t} = \sigma \left(\sum_{j \in \mathcal{N}_t(i)} \alpha_{ij,t} \mathbf{W}\mathbf{x}_{j,t} \right). \quad (38)$$

Multi-head attention is used to capture distinct modes of peer interaction:

$$\mathbf{h}'_{i,t} = \big\|_{m=1}^H \sigma \left(\sum_{j \in \mathcal{N}_t(i)} \alpha_{ij,t}^{(m)} \mathbf{W}^{(m)} \mathbf{x}_{j,t} \right). \quad (39)$$

In the implementation, the GAT output is followed by layer normalization, ELU activation, and dropout. Self-loops are supplied explicitly by the data pipeline, so the graph convolution layer is configured without automatic self-loop insertion.

5.4 Temporal Convolutional Mechanism

After spatial encoding, annual embeddings are stacked into a temporal tensor:

$$\mathbf{H} \in \mathbb{R}^{N \times C \times T}, \quad (40)$$

where C is the concatenated spatial embedding dimension and T is the number of annual snapshots. A causal one-dimensional convolution is then applied along the temporal dimension, following the standard temporal-convolution principle that predictions at time t should not depend on future states (Bai et al., 2018):

$$\mathbf{z}_{i,t} = \sigma \left(\sum_{k=0}^{K-1} \mathbf{W}_T^{(k)} \mathbf{h}'_{i,t-k} + \mathbf{b}_T \right). \quad (41)$$

The causal truncation ensures that the representation at year t depends only on current and historical graph states. The final output layer maps the temporal representation to two forward targets:

$$\hat{\mathbf{y}}_{i,t} = [\hat{R}_{i,t}, \hat{\sigma}_{i,t}]. \quad (42)$$

5.5 Masked Multi-Task Loss

Because the global universe contains firms that are not active in every year, the model uses a target-aware active mask $M_{t,i} \in \{0, 1\}$. A firm-year is active only when it has valid risk exposure data, valid financial features, and non-missing forward return and volatility labels. The training objective is the masked Huber loss averaged over active firm-years. Huber loss is used because it is less sensitive to extreme return observations than squared-error loss while remaining differentiable around zero (Huber, 1964):

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{1}{\sum_i M_{t,i}} \sum_{i=1}^N M_{t,i} \ell_{\delta}(\hat{\mathbf{y}}_{i,t}, \mathbf{y}_{i,t}), \quad (43)$$

where ℓ_{δ} is the Huber loss with $\delta = 1.0$. The first output dimension $\hat{R}_{i,t}$ denotes predicted one-year-ahead cumulative return, and the second output dimension $\hat{\sigma}_{i,t}$ denotes predicted one-year-ahead annualized volatility. Return and volatility are optimized jointly with equal weight, while evaluation reports target-specific RMSE, MAE, and Spearman rank correlation.

6 ST-GAT EXPERIMENTAL SETUP AND TENSOR CONSTRUCTION

This section consolidates the experimental setup for the ST-GAT analysis. Section 3 documents the data sources and economic transformations; the present section describes how those data are converted into supervised graph tensors, how the chronological split is imposed, how hyperparameters are selected, and how the strict identity baseline is defined. The separation is important: Section 3 answers what the data are, while this section answers how the data enter the forecasting experiment.

6.1 Node Feature Engineering and Train-Only Standardization

For each node v_i and year t , the feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^{12}$ consists of six firm-level features and six market or macro proxy features. The twelve inputs are Vol, Downside_Vol, MA_200_Ratio, Momentum_11M, Dollar_Volume, Log_Dollar_Volume, SP500_Ret, SP500_Vol, VIX_Avg, VIX_Change, TenY_Yield_Avg, and TenY_Yield_Change.

Feature preprocessing is fitted only on the training split. Training-year medians are used to impute missing active-node features, and training-year means and standard deviations are used for standardization. The same fitted statistics are then applied to validation and OOS test years. This prevents information from 2017 onward from entering the feature scaling process.

Table 7: Final ST-GAT node feature set ($F = 12$).

Feature Category	Specific Metrics	Frequency
Firm-specific price and risk	Vol; Downside_Vol; MA_200_Ratio; Momentum_11M	Annual window
Firm-specific liquidity	Dollar_Volume; Log_Dollar_Volume	Annual window
Market regime	SP500_Ret; SP500_Vol; VIX_Avg; VIX_Change	Annual window
Interest-rate proxy	TenY_Yield_Avg; TenY_Yield_Change	Annual window

6.2 Feature-Target Alignment

The data pipeline follows a July-to-June alignment. For risk year t , financial and market features are measured from July 1 of year t to June 30 of year $t + 1$. Forward targets are measured from July 1 of year $t + 1$ to June 30 of year $t + 2$. This matches the annual disclosure cycle while ensuring that the prediction target is strictly forward-looking.

The archived empirical panel contains 2021–2024 OOS rows. Because the risk-year 2024 target window extends to 2026-06-30, those rows should be treated as provisional until the window fully closes; this caveat affects the final archival refresh but not the relative comparison between ST-GAT and identity baselines within the same exported panel.

6.3 Graph Snapshot Tensor Structure

For each year t , the graph builder produces:

$$\mathbf{X}_t \in \mathbb{R}^{N \times F}, \quad (44)$$

$$\text{edge_index}_t \in \mathbb{N}^{2 \times E_t}, \quad (45)$$

$$\text{edge_attr}_t \in \mathbb{R}^{E_t}, \quad (46)$$

$$\mathbf{Y}_t \in \mathbb{R}^{N \times 2}, \quad (47)$$

$$\mathbf{M}_t \in \{0, 1\}^N. \quad (48)$$

The complete supervised panel is then represented as:

$$\mathbf{Y} \in \mathbb{R}^{N \times T \times 2}, \quad \mathbf{M} \in \{0, 1\}^{T \times N}. \quad (49)$$

Unlike a random mini-batch design, training is performed on the full chronological graph panel with masked losses. This is appropriate because the graph topology and temporal convolution depend on ordered annual snapshots.

6.4 Reproducibility and Output Verification

An independent validation audit checks source-table availability, CSV schemas, required columns, duplicate firm-years, chronological split labels, train-only preprocessing statistics, graph self-loop logic, identity-baseline edges, portfolio timing, transaction-cost conversion, and deterministic random-number generation. The validation result for the empirical evidence audited in this study is 35 passed checks, 7 warnings, and 0 failures. The warnings are transparent scope and research caveats: optional multi-seed, DAV, and taxonomy-interim evidence is not used as headline evidence; four missing feature values in the enhanced financial matrix are imputed downstream using training-year medians; and the exported panel includes 2024 rows whose final archival target window closes on 2026-06-30. Table 8 summarizes the main empirical sources behind the reported results.

Table 8: Output verification map for major empirical claims.

Claim Area	Primary Evidence	Generation Procedure	Status
Macro ST-GAT OOS	Macro OOS goodness-of-fit table	Macro forecasting stage	Verified for review panel
Meso ST-GAT OOS	Meso OOS goodness-of-fit table	Meso forecasting stage	Verified for review panel
Robust OOS comparison	Robust OOS evaluation table	Paired OOS diagnostic stage	Verified for review panel
Graph density	Macro/meso graph diagnostic tables	Graph construction and forecasting stages	Verified
Portfolio summary and spreads	Portfolio summary and spread tables	Portfolio backtest stage	Verified
Portfolio robust inference	Portfolio robust-inference tables	Portfolio inference stage	Verified for review panel
DAV peak diagnostics	DAV peak summary table and audit note	Peak-event diagnostic stage	Figure-derived; verified
Taxonomy coverage	Taxonomy coverage figure	Underlying source table	Verified
DAV appendix	BIC/FMB/SAR plots and tables; source tables incomplete or absent	Econometric validation stages	Deferred validation

6.5 Forecast Experiment Design and Algorithmic Tuning

The empirical design evaluates whether textual risk topology improves one-year-ahead return and volatility prediction under strict chronological splitting. Random cross-sectional splits are

avoided because they would leak future macro regimes into model selection.

6.5.1 Strict Chronological Data Splitting

The sample is split into three non-overlapping temporal regimes:

- **Training Set (2006–2016):** Used to fit model weights, feature standardization statistics, and median imputation values.
- **Validation Set (2017–2020):** Used for Optuna hyperparameter selection and early stopping.
- **Out-of-Sample Test Set (2021–2024):** Used only for final evaluation.

6.5.2 Dual-Level Spatial Ablation Framework

Three graph settings are evaluated:

1. **Macro-Level Topology:** Edges are constructed from broad first-level risk category exposure vectors.
2. **Meso-Level Topology:** Edges are constructed from granular second-level risk category exposure vectors.
3. **Identity Baseline:** All off-diagonal risk edges are removed, leaving only self-loops.

This design tests both whether textual peer topology improves forecasting and whether semantic resolution affects the quality of that topology.

6.5.3 Algorithmic Tuning and Optimization

Macro and Meso graphs have different densities and information bandwidths, so they are tuned independently. We use Optuna with a Tree-structured Parzen Estimator sampler and a Median Pruner, following modern automated hyperparameter optimization practice (Akiba et al., 2019). Each study runs 50 trials. The objective minimizes validation Huber loss over 2017–2020, with maximum 200 epochs per trial, patience of 25 epochs, gradient clipping at 1.0, and early pruning of weak trials. The final forecast models are trained with maximum 300 epochs, patience of 40 epochs, and restoration of the best validation checkpoint.

Table 9: Optuna search space and final hyperparameters.

Hyperparameter	Search Space	Macro ST-GAT	Meso ST-GAT
Similarity Threshold (τ)	Uniform: [0.75, 0.95]	0.8128	0.8292
Attention Heads (H)	Categorical: {1, 2, 4, 8}	1	2
Spatial Dimension	Categorical: {8, 16, 32}	16	8
Dropout Rate (p)	Uniform: [0.10, 0.50]	0.3688	0.4979
Learning Rate (η)	Log-uniform: [10^{-4} , 10^{-2}]	6.69×10^{-3}	8.08×10^{-3}
Weight Decay	Log-uniform: [10^{-6} , 10^{-3}]	1.98×10^{-5}	5.83×10^{-5}
Best Validation Huber Loss	–	0.0378	0.0347

The numerical best-parameter table above is the auditable hyperparameter record for this study. For taxonomy-variant robustness, the same validation-selected forecast procedure is applied to each completed full-window taxonomy path. This preserves comparability: when taxonomy hyperparameters change, the experiment asks whether the induced graph topology improves forecasting relative to its matched identity baseline, rather than whether an entirely different neural-network search procedure happens to perform better.

7 OUT-OF-SAMPLE PERFORMANCE AND EMPIRICAL RESULTS

This section reports OOS predictive performance on the 2021–2024 panel under completed full-window taxonomy specifications. The empirical analysis includes paired Macro and Meso downstream evidence for two taxonomy paths: the default dynamic taxonomy with $\theta = 0.45$, and the stricter semantic-deviation variant with $\theta = 0.40$. For each taxonomy path and each semantic level, ST-GAT is compared with a matched identity baseline that uses the same node features, targets, split, and model family but removes all off-diagonal textual peer links. This design ensures that the estimated graph contribution is not confounded with feature engineering, target construction, or the choice to report only one taxonomy level.

Performance is measured by RMSE, MAE, and Spearman rank correlation. RMSE and MAE evaluate point accuracy, while Spearman rank correlation evaluates cross-sectional ordering. The distinction is central in quantitative finance: a model can reduce forecast error without producing a superior trading rank, and a model can produce useful rankings even when point forecasts remain noisy.

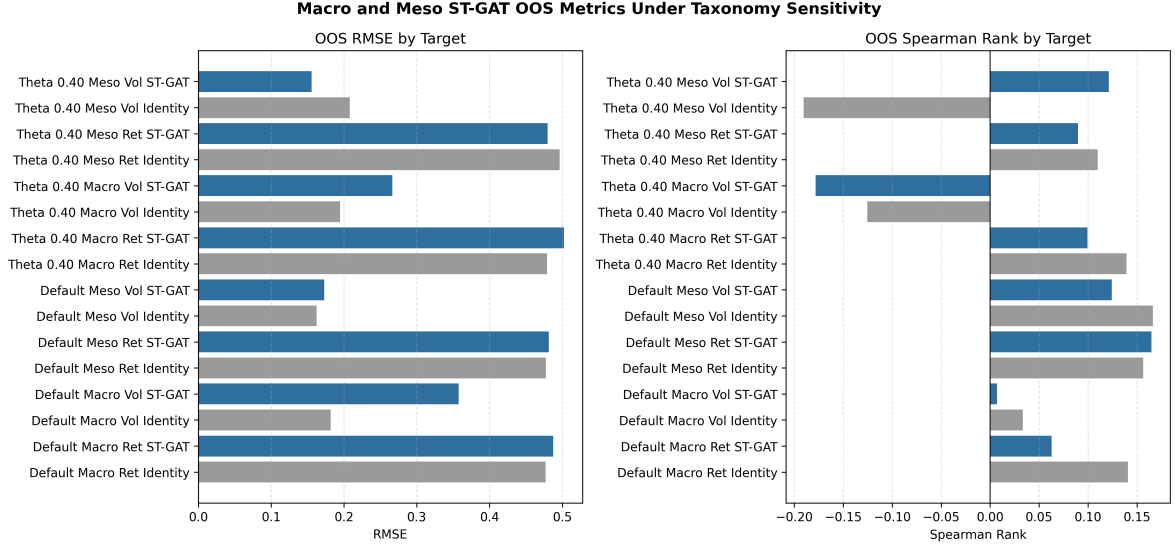


Figure 10: Empirical OOS metric comparison for Macro and Meso ST-GAT specifications under the default and $\theta = 0.40$ taxonomy paths. Lower RMSE is better; higher Spearman rank correlation is better.

7.1 Paired Macro–Meso Sensitivity Results

Table 10 is the direct Section 7 counterpart to the full-window taxonomy sensitivity design in Section 4. For every completed taxonomy path, the experiment is run at both semantic levels. This comparison is important because Macro graphs and Meso graphs fail in different ways. Macro categories create many peer links, but they can over-connect firms that share broad disclosure language for economically different reasons. Meso categories preserve semantic specificity, but they can become too sparse if the taxonomy fragments excessively.

Table 10: Paired Macro and Meso OOS sensitivity results. Positive improvement favors ST-GAT over its matched identity baseline.

Taxonomy	Level	Avg. RMSE Imp.	Avg. MAE Imp.	Vol. RMSE Imp.	Mean OOS Density
Default $\theta = 0.45$	Macro	-0.0933	-0.0640	-0.1760	40.756%
Default $\theta = 0.45$	Meso	-0.0073	-0.0021	-0.0106	0.114%
$\theta = 0.40$	Macro	-0.0478	-0.0385	-0.0719	74.743%
$\theta = 0.40$	Meso	0.0347	0.0535	0.0527	0.488%

The paired comparison is sharp. Macro graphs are far denser than Meso graphs, with mean OOS directed-edge densities of 40.756% under the default taxonomy and 74.743% under $\theta = 0.40$. Yet both Macro ST-GAT runs underperform their identity baselines on averaged RMSE and MAE. This indicates that broad category coverage alone is not enough; excessive graph density can transmit noisy peer information. The best-tested specification is therefore the $\theta = 0.40$ Meso graph: it is denser than the default Meso graph but still selective enough to avoid the overconnected Macro topology.

7.2 Robust Baseline Comparison

To avoid over-interpreting point estimates, the OOS comparison uses paired firm-year errors, year-block bootstrap intervals, and annual-block loss-difference tests. Because the OOS window contains only four risk years, these tests are diagnostic rather than definitive. The relevant estimand is not the unconditional accuracy of ST-GAT, but the paired improvement of ST-GAT over the identity baseline under the same taxonomy path.

Table 11: Robust OOS improvement diagnostics versus identity baseline. Positive values favor ST-GAT.

Variant	Target	RMSE Improvement	MAE Improvement	Spearman Improvement
Default	Return	-0.0041 [-0.0152, 0.0078]	0.0072 [-0.0142, 0.0315]	0.0086 [-0.0625, 0.0706]
Default	Volatility	-0.0106 [-0.0333, 0.0153]	-0.0115 [-0.0343, 0.0106]	-0.0418 [-0.1616, 0.3194]
$\theta = 0.40$	Return	0.0167 [-0.0053, 0.0427]	0.0552 [0.0221, 0.0854]	-0.0201 [-0.1649, 0.1310]
$\theta = 0.40$	Volatility	0.0527 [0.0265, 0.0802]	0.0517 [0.0249, 0.0776]	0.3118 [0.2382, 0.5365]

The robustness diagnostics materially change the interpretation. Under the default Meso taxonomy, the ST-GAT graph does not deliver a statistically stable OOS improvement over the identity baseline. Under the $\theta = 0.40$ Meso taxonomy, the strongest evidence is volatility forecasting: RMSE, MAE, and Spearman bootstrap intervals are all positive, and the annual-block squared-error test has a diagnostic p -value of 0.043. Return results remain weaker: the $\theta = 0.40$ Meso graph improves return RMSE and MAE, but rank correlation declines slightly.

The Macro-level robustness output reinforces this conclusion by negative example. The default Macro and $\theta = 0.40$ Macro graphs are much denser, yet both have negative averaged RMSE improvement relative to identity. Broad coverage therefore does not automatically imply useful information transfer. The defensible claim is conditional, not universal: graph topology is useful when the taxonomy evolution setting produces a semantic level that is neither too fragmented nor too broad, especially for volatility.

7.3 Seed-Stability Check

Neural-network results can be sensitive to random initialization. To avoid selecting a favorable seed ex post, the study defines a seed-stability protocol that reruns the final forecast stage using fixed hyperparameters and separates replication outputs from the headline forecast tables. Multi-seed replication is therefore part of the reproducibility design, while the reported numerical tables are restricted to the fully regenerated single-seed forecast panels used consistently in the portfolio stage.

7.4 Neural Network Convergence Dynamics

The final training stage uses validation early stopping and best-checkpoint restoration. Both full ST-GAT models and identity baselines are trained under the same chronological split, masked

Huber loss, and gradient clipping. Figure 11 compares validation-loss trajectories for the default and $\theta = 0.40$ taxonomy paths. The purpose of the figure is not to identify a visually pleasing training curve, but to check whether the reported OOS differences are consistent with validation behavior during training.

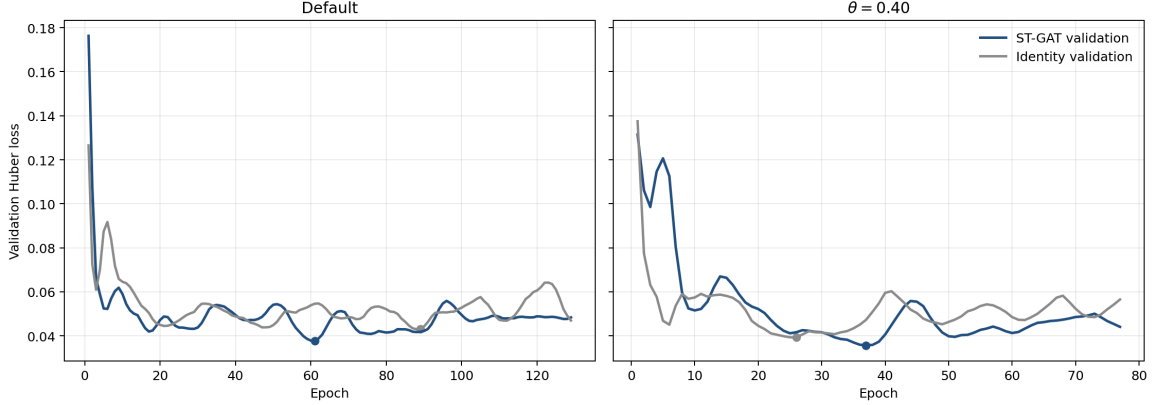


Figure 11: Validation Huber-loss convergence for ST-GAT and identity-baseline models under the default and $\theta = 0.40$ dynamic taxonomies. Dots mark the best validation checkpoints restored for OOS evaluation.

The $\theta = 0.40$ path exhibits a lower best validation loss for ST-GAT than the default path, while its identity baseline also improves. This is consistent with the interpretation that the stricter threshold changes the information geometry of the exposure matrix rather than merely benefiting the graph model. The key empirical question is therefore paired improvement: whether ST-GAT benefits more from the altered topology than the identity baseline benefits from the same target and feature panel.

7.5 Experiment 1: Default Meso Taxonomy

Table 12: OOS Performance: Default Meso ST-GAT vs. Identity Baseline (2021–2024)

Target	Model	RMSE ($\downarrow$)	MAE ($\downarrow$)	Spearman ($\uparrow$)
Returns	ST-GAT	0.4816	0.2813	0.1650
Returns	Identity Baseline	0.4775	0.2885	0.1564
Volatility	ST-GAT	0.1731	0.1523	0.1245
Volatility	Identity Baseline	0.1625	0.1408	0.1662
Averaged	ST-GAT	0.3274	0.2168	0.1447
Averaged	Identity Baseline	0.3200	0.2147	0.1613

The default Meso graph improves return MAE and return rank correlation, but it does not dominate the identity baseline on averaged RMSE, averaged MAE, or volatility rank correlation. This result is theoretically meaningful. A very granular taxonomy can improve interpretability

by separating risk themes, but it can also fragment firm exposure vectors. If two firms discuss economically related risks using categories that split into nearby but distinct Meso labels, their cosine similarity falls and the graph may fail to form a useful peer edge. The default taxonomy therefore provides a high-resolution semantic map but an extremely sparse OOS graph.

7.6 Experiment 2: Full-Window $\theta = 0.40$ Taxonomy Sensitivity

Table 13: OOS Performance: $\theta = 0.40$ Meso ST-GAT vs. Identity Baseline (2021–2024)

Target	Model	RMSE ($\downarrow$)	MAE ($\downarrow$)	Spearman ($\uparrow$)
Returns	ST-GAT	0.4797	0.2701	0.0898
Returns	Identity Baseline	0.4964	0.3252	0.1099
Volatility	ST-GAT	0.1555	0.1323	0.1212
Volatility	Identity Baseline	0.2082	0.1841	-0.1906
Averaged	ST-GAT	0.3176	0.2012	0.1055
Averaged	Identity Baseline	0.3523	0.2546	-0.0403

The $\theta = 0.40$ sensitivity run is stronger for forecast errors. It improves averaged RMSE and MAE, and the largest improvement appears in volatility forecasting, where the ST-GAT reduces RMSE from 0.2082 to 0.1555 and improves Spearman rank from -0.1906 to 0.1212. The mechanism is consistent with the taxonomy diagnostics in Section 4. The stricter threshold produces fewer final categories, reduces category fragmentation, and increases exposure overlap across firms. That creates a denser semantic peer graph, allowing the attention layer to aggregate more cross-sectional information.

The result also reveals the economic limit of the improvement. Return ranking remains weaker than the identity baseline, which indicates that the $\theta = 0.40$ topology is better at identifying shared risk-state information than at ordering expected winners and losers. Volatility is more naturally linked to common risk exposure and contagion, while realized returns contain a larger idiosyncratic component and are more sensitive to valuation, earnings surprises, and market timing.

7.7 Graph Density Diagnostics: Default versus $\theta = 0.40$ Topology

Graph diagnostics explain why taxonomy sensitivity matters. The default Meso graph is extremely sparse in the OOS period. Lowering the semantic-deviation threshold to $\theta = 0.40$ creates fewer final categories but a denser OOS peer topology, because more paragraphs are absorbed into existing categories and firm exposure vectors become less fragmented.

Table 14: OOS graph-density comparison between default and $\theta = 0.40$ Meso risk topologies.

Year	Default Edges	Default Density	$\theta = 0.40$ Edges	$\theta = 0.40$ Density
2021	32	0.061%	630	1.196%
2022	50	0.086%	198	0.342%
2023	96	0.154%	154	0.247%
2024	92	0.154%	100	0.167%

The $\theta = 0.40$ graph is especially denser in 2021, with 630 off-diagonal directed edges versus 32 for the default taxonomy. This helps explain why the sensitivity run can improve volatility forecast errors: it gives the ST-GAT more peer information to aggregate. It also warns that taxonomy hyperparameters affect both interpretability and model behavior.

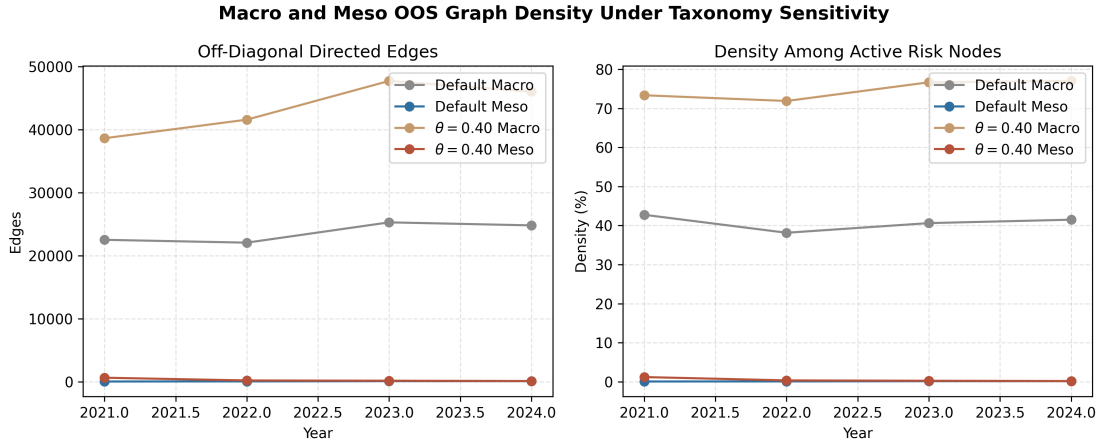


Figure 12: Empirical graph-density comparison for the OOS years, generated from the paired Macro and Meso graph-diagnostics CSV files under the default and $\theta = 0.40$ taxonomy paths.

Figure 13 summarizes the taxonomy-to-forecast propagation. Among the completed full-window variants, $\theta = 0.40$ is the better forecasting specification because it improves averaged RMSE and volatility RMSE relative to its identity baseline. It is not unambiguously the better investment specification, because its return-only long-short Sharpe is similar but its raw top-minus-bottom spread is smaller than under the default taxonomy. This divergence is an important empirical finding: the taxonomy that best supports volatility information transfer is not necessarily the taxonomy that maximizes return-ranking separation.

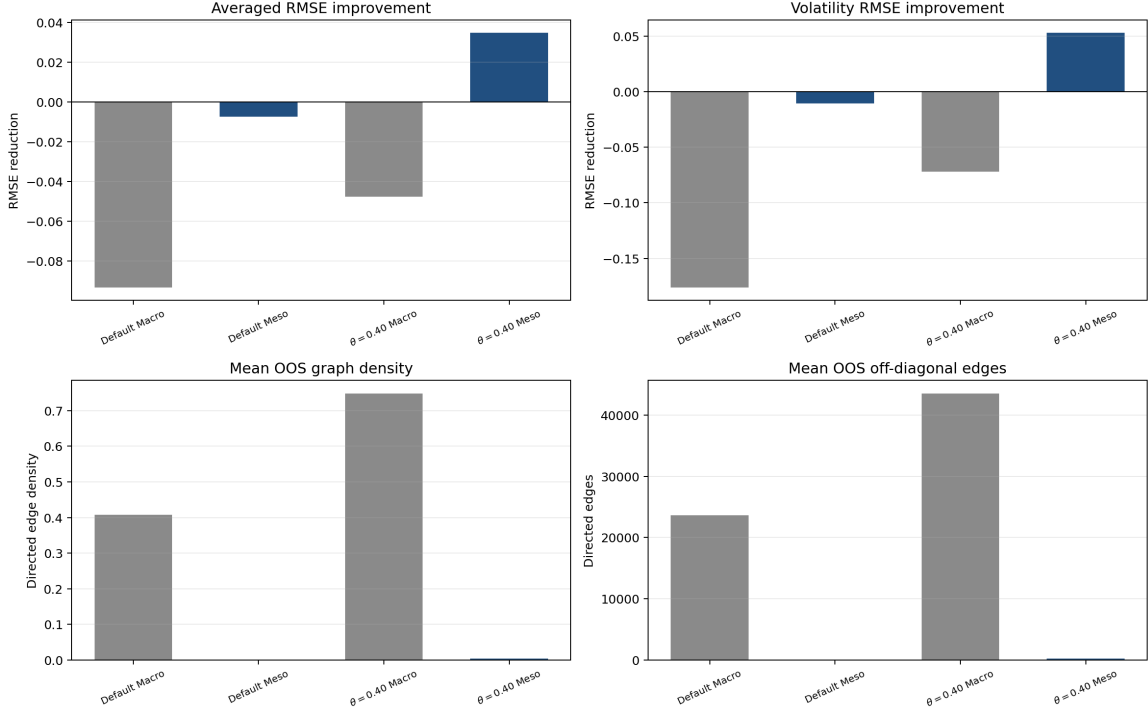


Figure 13: Downstream sensitivity dashboard for completed full-window taxonomy paths. Positive RMSE improvement indicates that ST-GAT outperforms its matched identity baseline. Graph density is averaged across 2021–2024 OOS years.

8 DYNAMIC PORTFOLIO BACKTESTING EXTENSION

The forecasting results show that Meso topology can matter, but also that the effect depends on taxonomy settings and target choice. A statistically better forecast does not automatically imply a tradable investment strategy. To evaluate whether the model produces economically meaningful signals, we extend the forecasting output into a monthly rebalanced portfolio backtest. This section formalizes the signal timing, portfolio construction rules, transaction-cost assumptions, score ablations, benchmark comparison, top-minus-bottom spread diagnostic, and downstream taxonomy sensitivity result.

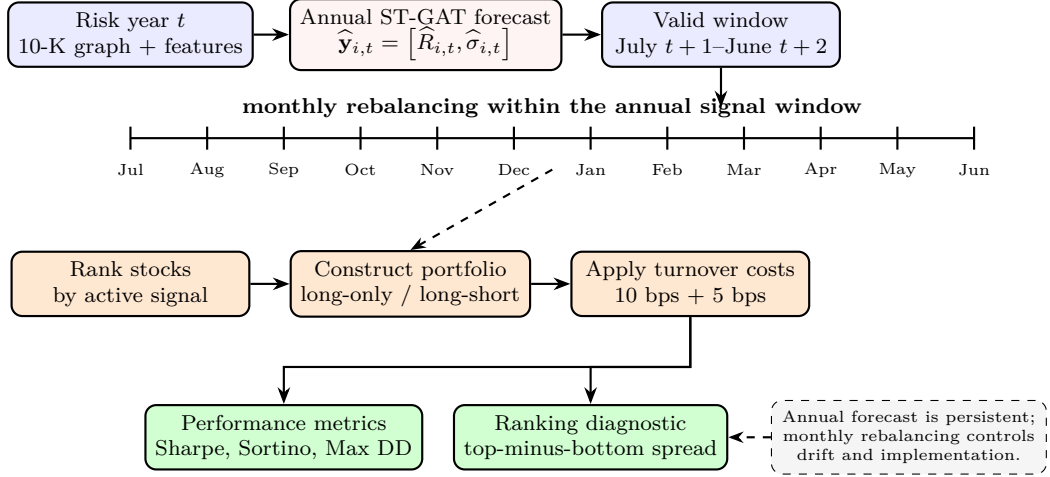


Figure 14: Portfolio signal timing. Annual ST-GAT predictions remain active over the July-to-June forecast window, while monthly rebalancing controls weight drift, turnover, and implementation costs.

8.1 Objective and Signal Choice

The portfolio extension is designed to answer whether the ST-GAT prediction panel can separate future winners from losers after accounting for realistic implementation frictions. The primary signal source is the Meso forecast panel because Meso exposures are the most semantically informative graph inputs. The identity-baseline predictions are retained as a portfolio benchmark, allowing the analysis to test whether textual peer topology improves allocation quality, not merely predictive loss.

For each firm-year (i, t) , the forecast panel contains two model outputs:

$$\hat{\mathbf{y}}_{i,t} = \left[\hat{R}_{i,t}, \hat{\sigma}_{i,t} \right], \quad (50)$$

where $\hat{R}_{i,t}$ is predicted forward cumulative return and $\hat{\sigma}_{i,t}$ is predicted forward annualized volatility. These predictions are generated for the same July-to-June target window used in the supervised learning problem.

8.2 Annual Signal Persistence and Monthly Rebalancing

The ST-GAT model produces annual forecasts rather than monthly forecasts. A prediction associated with risk year t is therefore treated as valid from July $t + 1$ through June $t + 2$. Within this 12-month forecast window, the signal ranking is held fixed unless a new annual prediction becomes available. Monthly rebalancing is nevertheless economically meaningful because it controls weight drift, refreshes investable eligibility based on available price data, realizes transaction costs, and maintains the intended long, short, and risk-budget exposures.

The timing convention is designed to avoid look-ahead bias. At month-end m , the strategy observes the active annual signal available at that date, forms target weights, incurs turnover-based transaction costs, and earns the return from month m to month $m + 1$. Thus, the rebalance

decision and realized holding-period return are separated by one monthly interval.

$$\mathbf{w}_m^{target} = f(\mathcal{S}_m, \mathcal{U}_m), \quad R_{p,m+1} = \sum_{i \in \mathcal{U}_m} w_{i,m}^{target} R_{i,m+1} - \text{Cost}_m, \quad (51)$$

where $\mathcal{S}_m$ is the active annual signal set and $\mathcal{U}_m$ is the investable universe with valid next-month return observations.

8.3 Score Definitions and Portfolio Ranking

Because the ST-GAT produces both return and volatility forecasts, we evaluate three alternative scoring rules. The first uses predicted return directly:

$$\text{Score}_{i,t}^{ret} = \hat{R}_{i,t}. \quad (52)$$

The second converts predictions into a risk-adjusted return score:

$$\text{Score}_{i,t}^{ret/vol} = \frac{\hat{R}_{i,t}}{\max(\hat{\sigma}_{i,t}, 0.10)}, \quad (53)$$

where the 10% volatility floor prevents unstable weights caused by very small predicted volatility values. The third score uses standardized return and volatility forecasts:

$$\text{Score}_{i,t}^{comp} = z(\hat{R}_{i,t}) - 0.5z(\hat{\sigma}_{i,t}). \quad (54)$$

This composite score rewards high predicted return while penalizing high predicted volatility, without imposing the stronger functional form of a return-to-volatility ratio.

8.4 Long-Only and Long-Short Portfolio Construction

For each score definition, two strategies are constructed. The first is a long-only top-quintile portfolio. At each rebalance date, stocks are ranked by the active score, and the top 20% of eligible firms are selected. Within the selected set, weights are scaled inversely by predicted volatility:

$$\tilde{w}_{i,m} = \frac{1/\max(\hat{\sigma}_{i,t}, 0.10)}{\sum_{j \in \mathcal{Q}_m^{top}} 1/\max(\hat{\sigma}_{j,t}, 0.10)}. \quad (55)$$

A 5% single-name cap is applied to reduce concentration risk, after which weights are renormalized to sum to one.

The second strategy is a long-short top-minus-bottom quintile portfolio. The top 20% of stocks by score are held long and the bottom 20% are held short. The long leg is allocated 50% gross exposure and the short leg is allocated -50% gross exposure:

$$\sum_{i \in \mathcal{Q}_m^{top}} w_{i,m} = 0.5, \quad \sum_{i \in \mathcal{Q}_m^{bottom}} w_{i,m} = -0.5. \quad (56)$$

This design creates an approximately market-neutral cross-sectional ranking test, while still allowing non-zero beta because no explicit beta-neutral optimization is imposed.

8.5 Transaction Costs, Slippage, and Turnover

Implementation frictions are modeled using one-way turnover. Let $\mathbf{w}_m^{target}$ denote the target weights at rebalance month m and $\mathbf{w}_m^{pre}$ denote the drifted weights immediately before rebalancing. Turnover is:

$$\text{Turnover}_m = \sum_i |w_{i,m}^{target} - w_{i,m}^{pre}|. \quad (57)$$

The total one-way implementation cost is set to 15 bps, consisting of 10 bps transaction cost and 5 bps bid-ask/slippage allowance:

$$\text{Cost}_m = 0.0015 \times \text{Turnover}_m. \quad (58)$$

Net portfolio return is therefore:

$$R_{p,m+1}^{net} = R_{p,m+1}^{gross} - \text{Cost}_m. \quad (59)$$

This treatment penalizes both initial portfolio formation and subsequent monthly rebalancing, making the backtest more conservative than a frictionless ranking exercise.

8.6 Evaluation Metrics

Portfolio performance is evaluated using total return, annualized return, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, hit rate, average monthly turnover, alpha and beta against SPY, information ratio, and correlation with the benchmark. The Sharpe ratio is the standard mean-variance reward-to-risk measure, while the Sortino ratio focuses on downside risk (Sharpe, 1966; Sortino and Price, 1994).

8.7 Portfolio Performance Results

Table 15 reports the headline strategies under the default dynamic Meso taxonomy. The strongest long-short ST-GAT strategy ranks stocks by predicted return only, rather than by predicted return divided by predicted volatility. This finding is important: although predicted volatility can improve the forecasting task, using volatility directly in the denominator can over-penalize high-return opportunities.

Table 15: Key portfolio backtest results for default dynamic Meso ST-GAT signals.

Strategy	Ann. Ret.	Ann. Vol.	Sharpe	Sortino	Max DD
GAT Long-Short Return-Only	7.31%	8.34%	0.88	1.52	-8.06%
GAT Long-Only Return-Only	23.55%	23.81%	0.99	2.04	-17.85%
SPY Benchmark	10.74%	16.56%	0.65	1.19	-23.93%

The GAT long-short return-only portfolio achieves 7.31% annualized return with 8.34% annualized volatility, producing a Sharpe ratio of 0.88 and Sortino ratio of 1.52. Its maximum drawdown is -8.06%, lower than SPY’s -23.93% over the same exported backtest window. Its beta against SPY is approximately 0.30, indicating that the strategy is not merely a one-for-one market exposure. The GAT long-only return-only portfolio generates 23.55% annualized return and a Sharpe ratio of 0.99, but its beta is 1.32 and correlation with SPY is 0.93; this is economically useful but less clean as a pure textual-risk ranking test than the long-short portfolio.

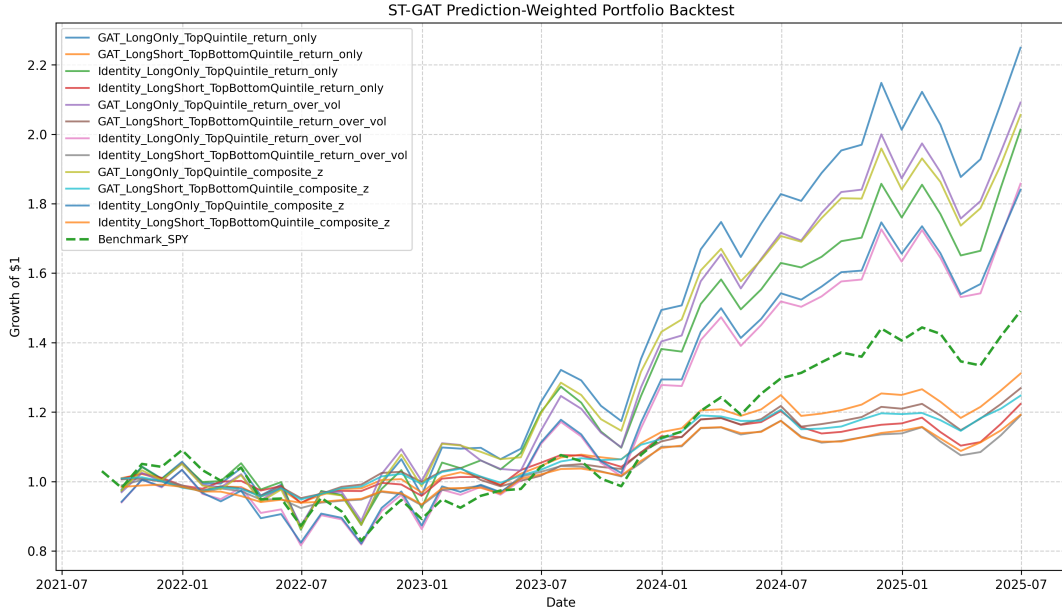


Figure 15: Monthly rebalanced ST-GAT and identity-baseline portfolio equity curves.

8.8 Score Ablation Interpretation

The score ablation shows that the ST-GAT return-only ranking is more effective than the return-over-volatility ranking. This does not contradict the volatility-forecasting improvement documented in Section 7. Rather, it indicates that the predicted volatility output is useful for understanding and controlling risk, but the most direct source of cross-sectional alpha is the predicted return ranking. The composite score performs between the return-only and return-over-volatility specifications, suggesting that moderate volatility penalization is preferable to mechanically dividing by predicted volatility.

This result also clarifies the difference between forecast evaluation and trading evaluation. RMSE and MAE measure point-forecast accuracy, while a portfolio primarily depends on the ordinal separation of expected winners and losers. Therefore, the portfolio section emphasizes Sharpe ratio, drawdown, beta, turnover, and top-minus-bottom spread in addition to predictive metrics.

8.9 Top-Bottom Spread Diagnostic

To directly test whether the predicted rankings separate future winners from losers, we compute the monthly average return of the top quintile minus the bottom quintile. This diagnostic is model-agnostic: it does not depend on inverse-volatility weights, portfolio caps, or transaction-cost assumptions. It simply asks whether stocks ranked highly by the signal outperform stocks ranked poorly by the signal in the next month.

Table 16: Top-minus-bottom spread diagnostics for return-only signals.

Signal	Annualized Spread	t -stat	p -value	Positive Spread Rate
ST-GAT Return-Only	16.26%	1.79	0.080	63.0%
Identity Return-Only	11.47%	1.30	0.199	63.0%

The ST-GAT return-only signal produces a 16.26% annualized top-minus-bottom spread, with $t = 1.79$ and $p = 0.080$ before adjustment for multiple score/model comparisons. The spread is positive in 63.0% of evaluated months. The identity baseline return-only signal produces a smaller 11.47% annualized spread with $t = 1.30$ and $p = 0.199$.

A robustness addendum recomputes spread inference using HAC/Newey-West standard errors with three lags and Benjamini-Hochberg q -values across the six model/score spread tests. Under this diagnostic, the default ST-GAT return-only spread has HAC $t = 2.11$, $p = 0.034$, and $q = 0.076$. This supports a nominal ranking signal but does not survive a strict 5% Benjamini-Hochberg threshold. The evidence should therefore be read as academic backtest evidence, not as a live-trading claim.

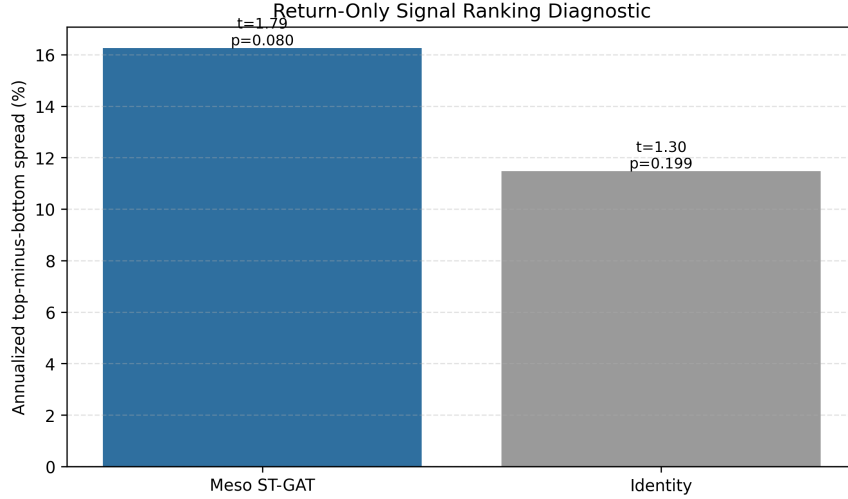


Figure 16: Return-only top-minus-bottom spread comparison for ST-GAT and identity-baseline signals. The reported p -values are nominal and not adjusted for multiple score/model comparisons.

8.10 Portfolio Sensitivity and Economic Interpretation

Table 17 compares the return-only long-short strategy across the completed full-window taxonomy paths. The table reports both realized portfolio performance and the model-agnostic top-minus-bottom spread. This distinction matters because a long-short portfolio includes inverse-volatility sizing, caps, turnover, and transaction costs, while the spread diagnostic isolates ranking quality.

Table 17: Return-only long-short portfolio sensitivity across completed taxonomy paths.

Taxonomy	Model	Ann. Ret.	Sharpe	Max DD	Spread	HAC p / BH q
Default $\theta = 0.45$	ST-GAT	7.31%	0.88	-8.06%	16.26%	0.034 / 0.076
Default $\theta = 0.45$	Identity	5.40%	0.62	-8.29%	11.47%	0.133 / 0.199
$\theta = 0.40$	ST-GAT	4.41%	0.92	-3.24%	9.33%	0.007 / 0.044
$\theta = 0.40$	Identity	6.00%	0.79	-6.98%	11.68%	0.092 / 0.170

The $\theta = 0.40$ taxonomy improves ST-GAT forecast errors, but its portfolio results are more mixed. The ST-GAT return-only long-short strategy earns 4.41% annualized return with 4.80% annualized volatility, a Sharpe ratio of 0.92, maximum drawdown of -3.24%, and beta of 0.13. Its top-minus-bottom spread is 9.33% annualized. By contrast, the matched identity-baseline return-only spread is 11.68% annualized. HAC inference still gives the ST-GAT return-only spread a positive diagnostic result ($t = 2.68$, $p = 0.007$, $q = 0.044$), but the lower raw spread relative to the identity baseline means the variant should not be interpreted as a stronger portfolio result.

This mismatch between forecast and portfolio sensitivity is economically informative. Forecast RMSE rewards point accuracy, while the long-short portfolio rewards ordinal separation of

winners and losers. The $\theta = 0.40$ graph appears useful for volatility and point-error reduction because the coarser taxonomy produces denser peer neighborhoods. However, return ranking is a sharper cross-sectional task. A taxonomy that groups firms into smoother exposure clusters can reduce volatility noise while simultaneously weakening the distinction between firms with similar risk language but different expected-return implications. In other words, the best taxonomy for volatility information transfer need not be the best taxonomy for alpha ranking.

The default taxonomy has the opposite strength. It is more fragmented and therefore weaker on averaged forecast-error metrics, but its higher semantic specificity produces the largest raw return-only top-minus-bottom spread. The evidence therefore suggests a taxonomy-resolution trade-off: coarser dynamic taxonomies may improve graph connectivity and volatility forecasting, while more granular taxonomies may retain sharper cross-sectional return information when enough peer links survive.

The portfolio extension supports three conclusions within an academic backtest setting. First, the default Meso ST-GAT signal is economically suggestive but not definitive. Second, the best-performing ST-GAT allocation uses predicted return directly, showing that the strongest ranking signal is embedded in the return forecast rather than the return-volatility ratio. Third, monthly rebalancing is compatible with annual predictions when the signal is interpreted as a persistent 12-month forecast and rebalancing is used for risk control and execution rather than signal refresh. These findings motivate further testing of LLM-derived semantic risk topology as an input to systematic equity selection, but they should not be interpreted as live-trading evidence.

9 LIMITATIONS

This study identifies several structural and methodological limitations within the proposed Spatio-Temporal Graph Attention Network, the underlying NLP pipeline, and the portfolio extension. First, there is an inherent frequency mismatch between annual 10-K risk disclosures and continuous market dynamics. In the core ST-GAT architecture, this is handled by representing textual risk as an annual cross-sectional graph snapshot. This design is appropriate for one-year-ahead return and volatility forecasting, but it cannot capture intra-year fundamental shocks unless additional higher-frequency textual or event data are incorporated.

Second, the supplemental DAV experiment illustrates the difficulty of mapping annual risk disclosures into daily volatility models. The full BIC horse-race table is currently empty and therefore not treated as verified evidence. The separate DAV peak-event figure set is more informative but still mixed: among 93 figure-derived diagnostics, only 20 show positive peak accuracy improvements and the mean improvement is -2.40%. This limitation motivates the graph-based cross-sectional architecture and cautions against interpreting annual textual risk scores as high-frequency volatility-trading signals.

Third, the predictive efficacy of the spatial graph topology is bounded by disclosure quality. Firms with boilerplate or near-zero-drift disclosures can introduce persistent but weakly informative

links into the adjacency matrix. The strict identity-matrix ablation and the Macro-versus-Meso comparison partially address this issue, but a future implementation should explicitly filter low-drift firms or down-weight stale disclosures when constructing graph topology.

Fourth, the dynamic taxonomy framework is sensitive to semantic-deviation threshold choice. The full-window sensitivity design covers alternative Macro, Meso, UMAP, and threshold settings, and the empirical manuscript reports complete downstream evidence for the default $\theta = 0.45$ and stricter $\theta = 0.40$ paths. The remaining grid points require the same long-horizon LLM-assisted evolution procedure before they can be interpreted as empirical results. There is also residual risk of LLM-induced semantic drift when categories are repeatedly added, merged, or reorganized.

Fifth, secondary validations such as the Spatial Autoregressive network analysis remain constrained by extreme matrix sparsity and small sample sizes. This highlights the difficulty of modeling continuous market dynamics strictly through annualized semantic linkages.

Sixth, the reported portfolio spread p -values are nominal. Because multiple score definitions and model variants are evaluated, a stricter confirmatory study should adjust for multiple testing or pre-register a single primary signal before making formal inference.

Seventh, the current validation suite identifies four missing feature values in the enhanced financial matrix. The graph builder handles these through training-year median imputation, which preserves train-only preprocessing, but the missingness should remain documented.

Finally, the portfolio backtest remains an academic backtest rather than a live-trading system. Transaction costs and slippage are included, but borrow constraints, short-sale availability, execution delay, corporate actions, liquidity constraints, and a fully survivorship-bias-free reconstruction of the S&P 500 universe remain only partially addressed. These limitations should be resolved before interpreting the strategy as deployable capital.

10 CONCLUSION & NEXT STEPS

This study develops a comprehensive, LLM-driven framework for transforming lengthy and unstructured SEC Item 1A risk disclosures into quantitative forecasting and portfolio-construction signals. The pipeline extracts risk paragraphs from 10-K filings, maps them into a hierarchical macro-meso taxonomy, converts firm-year disclosures into normalized risk exposure vectors, and uses those vectors to construct dynamic semantic peer graphs.

The central modeling contribution is a topology-only Spatio-Temporal Graph Attention Network with a strict identity-matrix ablation. The full model uses $\mathbf{A}_t^{full} = \mathbf{I} + \mathbf{A}_t^{risk}$, while the baseline uses $\mathbf{A}_t^{base} = \mathbf{I}$. This design isolates the incremental value of textual peer topology beyond each equity’s own financial features. In the 2021–2024 OOS panel, the default Meso taxonomy produces mixed results: return MAE and return rank correlation improve, but averaged RMSE and volatility rank do not. The $\theta = 0.40$ Meso sensitivity taxonomy produces stronger forecast-error evidence, improving averaged RMSE from 0.3523 to 0.3176 and averaged MAE from 0.2546

to 0.2012, with the largest gain in volatility RMSE. Macro-level paired experiments are much denser but perform worse than identity baselines, so the evidence favors selective Meso topology rather than broad Macro peer aggregation.

Graph diagnostics further clarify why taxonomy sensitivity matters. The default Meso graph retains 32–96 OOS off-diagonal directed edges per year, while the $\theta = 0.40$ Meso variant retains 100–630. Macro graphs retain tens of thousands of off-diagonal directed edges over the same OOS years, which is too broad to improve forecast errors. This change in graph density explains why taxonomy evolution settings must be validated downstream rather than treated as cosmetic NLP parameters.

The portfolio extension demonstrates that the Meso ST-GAT forecasts are economically suggestive but sensitive. Annual predictions are held fixed over their July-to-June forecast window and portfolios are rebalanced monthly to control drift, turnover, and implementation costs. Under the default taxonomy, the return-only long-short strategy achieves a Sharpe ratio of 0.88, Sortino ratio of 1.52, maximum drawdown of -8.06%, and beta of 0.30 against SPY. Its 16.26% annualized top-minus-bottom spread has conventional $p = 0.080$ and HAC/Newey-West $p = 0.034$ before multiple-testing adjustment. Under $\theta = 0.40$, forecast errors improve but the portfolio ranking advantage weakens relative to the identity baseline. The correct conclusion is that semantic risk topology is promising, but its investment interpretation requires taxonomy-sensitivity controls.

Supplemental DAV, Fama-MacBeth, and SAR analyses provide additional economic context but are not treated as headline evidence unless their source tables are archived and auditable. The DAV peak diagnostics support only a narrow, event-local interpretation and do not establish broad DAV dominance. The central verified result is the strict identity-matrix ablation plus the full-window sensitivity workflow showing where granular textual-risk topology helps, where it fails to dominate, and how taxonomy choices propagate into forecast and portfolio outcomes.

Future work should extend the framework in four directions. First, historical constituent reconstruction and delisting-aware price histories should be improved to further mitigate survivorship bias. Second, disclosure staleness and boilerplate intensity should be integrated directly into edge weighting or edge filtering. Third, monthly or quarterly textual/event signals could be added to support genuinely higher-frequency forecasts. Finally, the portfolio engine should be expanded to include borrow constraints, beta-neutral optimization, liquidity-aware execution, and transaction-cost stress testing.

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A Formula Compendium and Evaluation Metrics

This appendix collects supplementary equations used across the report. They are placed here to preserve main-text readability while keeping the methodology fully explicit.

A.1 Encoding and Embedding Examples

Before clustering, each Item 1A paragraph is converted into a context string,

$$x_i = \text{Heading}_i \oplus \text{Subheading}_i \oplus \text{RawText}_i,$$

and then mapped by the sentence-transformer encoder into a normalized vector $\mathbf{e}_i \in \mathbb{R}^{384}$. Individual coordinates are not interpreted as standalone economic variables; they are useful because their direction permits cosine-similarity comparisons between pieces of disclosure text.

A familiar embedding analogy is:

$$\mathbf{v}(\text{king}) - \mathbf{v}(\text{man}) + \mathbf{v}(\text{woman}) \approx \mathbf{v}(\text{queen}). \quad (60)$$

The thesis uses the same geometric intuition for risk language, but at the paragraph level rather than the single-word level. For example, a vector describing “pandemic travel restrictions” should lie closer to one describing “quarantine and casino-resort interruption” than to one describing “cybersecurity breach,” even when the exact words differ.

Table 18: Illustrative encoding examples using `all-MiniLM-L6-v2`. Only the first four normalized embedding coordinates are shown; the actual vector has 384 dimensions.

ID	Simplified encoded context	First four embedding coordinates
A	Integrated resort operations may be disrupted by infectious disease outbreaks, travel restrictions, reduced visitation, and temporary property closures.	(0.102, −0.012, 0.040, 0.046)
B	Casino operations in Macao depend on maintaining gaming concessions, complying with regulation, and satisfying licensing requirements.	(0.064, 0.035, −0.013, −0.072)
C	A failure of information systems, unauthorized access, or a data breach could disrupt operations and harm customer trust.	(−0.029, 0.001, 0.006, −0.044)

The economic point is directional similarity, not the meaning of coordinate 1 or coordinate 2. For example, a paraphrase of Example A that refers to pandemics, quarantines, border closures, and casino-resort interruption has cosine similarity 0.661 with Example A, but only 0.222 with the concession/licensing example and 0.278 with the cybersecurity example. This is why the pipeline can cluster paragraphs that use different wording but describe the same underlying risk theme.

A.2 Forecast Error and Ranking Metrics

For n evaluated firm-years with observed value y_i and prediction $\hat{y}_i$, the mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE) are:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (61)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (62)$$

Spearman rank correlation is the Pearson correlation between ranks:

$$\rho_S = \text{Corr}(\text{rank}(y_i), \text{rank}(\hat{y}_i)). \quad (63)$$

If an R^2 diagnostic is computed in future extensions, it is:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (64)$$

A.3 Graph Diagnostics

For an active graph with N_t^{act} active nodes and E_t^{risk} off-diagonal directed risk edges, density and degree statistics are:

$$\text{Density}_t = \frac{E_t^{risk}}{N_t^{act}(N_t^{act} - 1)}, \quad d_{i,t}^{out} = \sum_{j \neq i} A_{ij,t}^{risk}, \quad \bar{d}_t^{out} = \frac{1}{N_t^{act}} \sum_i d_{i,t}^{out}. \quad (65)$$

The identity baseline has no off-diagonal risk links:

$$E_{t,base}^{risk} = 0, \quad \mathbf{A}_t^{base} = \mathbf{I}_N. \quad (66)$$

A.4 Loss and Regularization

The scalar Huber loss for residual $u = y - \hat{y}$ is:

$$\ell_\delta(u) = \begin{cases} \frac{1}{2}u^2, & |u| \leq \delta, \\ \delta \left(|u| - \frac{1}{2}\delta \right), & |u| > \delta. \end{cases} \quad (67)$$

With model parameters Θ and weight decay λ , the regularized training objective is:

$$\mathcal{J}(\Theta) = \mathcal{L}_{Huber}(\Theta) + \lambda \|\Theta\|_2^2. \quad (68)$$

A.5 Portfolio Metrics

For monthly net portfolio returns $R_{p,m}$ and annualization factor $A = 12$, cumulative and annualized returns are:

$$R_p^{cum} = \prod_{m=1}^M (1 + R_{p,m}) - 1, \quad R_p^{ann} = (1 + R_p^{cum})^{A/M} - 1. \quad (69)$$

Annualized volatility, Sharpe ratio, and Sortino ratio are:

$$\sigma_p^{ann} = \sqrt{A} \text{sd}(R_{p,m}), \quad \text{Sharpe} = \frac{R_p^{ann} - R_f^{ann}}{\sigma_p^{ann}}, \quad (70)$$

$$\text{Sortino} = \frac{R_p^{ann} - R_f^{ann}}{\sqrt{A} \text{sd}(\min(R_{p,m} - R_f/A, 0))}. \quad (71)$$

The report uses $R_f = 0$ for the academic backtest unless otherwise stated. Maximum drawdown is computed from the equity curve $C_m = \prod_{s=1}^m (1 + R_{p,s})$:

$$\text{MDD} = \min_m \left(\frac{C_m}{\max_{s \leq m} C_s} - 1 \right). \quad (72)$$

Benchmark beta against SPY is:

$$\beta_p = \frac{\text{Cov}(R_{p,m}, R_{b,m})}{\text{Var}(R_{b,m})}. \quad (73)$$

Portfolio return, turnover, and transaction-cost adjustment are:

$$R_{p,m+1}^{gross} = \sum_i w_{i,m}^{target} R_{i,m+1}, \quad \text{Turnover}_m = \sum_i |w_{i,m}^{target} - w_{i,m}^{pre}|, \quad (74)$$

$$R_{p,m+1}^{net} = R_{p,m+1}^{gross} - c \text{Turnover}_m, \quad c = 0.0015. \quad (75)$$

A.6 Statistical Validation

For a statistic $\hat{\theta}$ estimated from annual or monthly blocks, a percentile bootstrap confidence interval with B resamples is:

$$CI_{1-\alpha} = \left[q_{\alpha/2} \left(\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)} \right), q_{1-\alpha/2} \left(\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)} \right) \right]. \quad (76)$$

For a time-series mean $\bar{x}$ with autocovariance estimates $\hat{\Gamma}_\ell$ and Newey-West lag L , the HAC variance estimator is:

$$\widehat{\text{Var}}_{NW}(\bar{x}) = \frac{1}{T} \left[\hat{\Gamma}_0 + 2 \sum_{\ell=1}^L \left(1 - \frac{\ell}{L+1} \right) \hat{\Gamma}_\ell \right]. \quad (77)$$

In the Fama-MacBeth setting, monthly cross-sectional premiums $\hat{\lambda}_{k,t}$ are averaged as:

$$\bar{\lambda}_k = \frac{1}{T} \sum_{t=1}^T \hat{\lambda}_{k,t}, \quad t_{NW}(\bar{\lambda}_k) = \frac{\bar{\lambda}_k}{\sqrt{\widehat{\text{Var}}_{NW}(\bar{\lambda}_k)}}. \quad (78)$$

B Econometric Baseline Specification (DAV Model)

To establish a traditional econometric comparison for textual risk integration, this study specifies a Disclosure Augmented Volatility (DAV) model. The DAV framework extends a standard Exponential GARCH (EGARCH) model by introducing textual risk scores as exogenous variables (EGARCH-X). This appendix is supplemental. The aggregate DAV benchmark table is incomplete in the present archival evidence set, so the final report does not make aggregate DAV BIC-improvement claims.

B.1 Data Sources and Preprocessing

The model evaluates the 2020–2025 temporal regime. Daily adjusted closing prices were sourced via Yahoo Finance to calculate realized continuous returns (r_t). The textual risk signals utilized the granular Meso-level risk matrices derived from the 10-K filings. To align the annual reporting frequency of 10-Ks with daily trading data, the annual risk scores were forward-filled, creating a daily step-function vector for the risk level (S_t) and a discrete difference vector for the risk shock (dS_t).

B.2 Mathematical Formulation

The DAV model is specified as an EGARCH(1,1)-X process. Unlike standard GARCH, the EGARCH framework models the natural logarithm of the conditional variance ($\ln(h_t)$), ensuring positive volatility without enforcing artificial non-negativity constraints on the coefficients. It also captures the asymmetric “leverage effect” of negative returns.

The textual risk factors are embedded directly into the conditional variance equation:

$$\ln(h_t) = \omega + \beta \ln(h_{t-1}) + \alpha(|z_{t-1}| - \mathbf{E}[|z|]) + \gamma z_{t-1} + \delta_1 S_t + \delta_2 dS_t \quad (79)$$

Where:

- h_t is the conditional variance at time t .
- z_{t-1} is the standardized residual $\frac{\epsilon_{t-1}}{\sqrt{h_{t-1}}}$.
- γ captures the asymmetric leverage effect of return shocks.
- δ_1 measures the persistent “Level Effect” of the firm’s textual risk score.
- δ_2 captures the transient “Shock Effect” representing year-over-year changes in disclosure.

The models were optimized using the L-BFGS-B algorithm to minimize the negative log-likelihood of the Gaussian distribution. Model selection was governed by the Bayesian Information Criterion (BIC), which severely penalizes models for adding uninformative parameters.

B.3 Results and Discussion

The original full-sample DAV horse-race is designed to compare GARCH, EGARCH, and DAV/EGARCH-X by likelihood-based criteria such as BIC. Because the aggregate benchmark table is incomplete, the final report omits case-specific BIC-improvement values and treats the full benchmark as a deferred validation task.

The peak-event diagnostic is separate from the full BIC benchmark table. Each diagnostic figure compares a GARCH baseline with a text-augmented DAV curve around the top four distinct volatility events for one ticker–risk pair. Because the peak-analysis evidence consists of figures rather than the original estimation table, the diagnostic is summarized from figure titles and treated as event-local evidence only.

Table 19: DAV peak-event diagnostic summary derived from 93 figures.

Diagnostic	Value
Peak-event figures scanned	93
Figure-title improvements parsed	93
Positive DAV peak improvements	20
Negative DAV peak improvements	73
Mean peak accuracy improvement	-2.40%
Median peak accuracy improvement	-1.91%
Standard deviation	2.95 percentage points
Range	-11.74% to 3.03%

Table 20: Largest positive DAV peak-event improvements in the available figure set.

Ticker	Macro Risk	Meso Risk	Improvement
SHW	Global Economic & Geopolitical Risk	International Operations and Market-Specific Risks	3.03%
NOW	Stock Price Volatility	Market and Performance Expectations Volatility	2.69%
WFC	Cybersecurity & Data Protection	Cybersecurity Incident & Breach Risk	2.36%
KO	Supply Chain & Operations	Third-Party Dependency and Outsourcing Risks	1.99%
META	Competitive & Market Risk	Disruption of Traditional Business Models	1.61%

Interpretation. The peak-event evidence weakens, rather than strengthens, any broad claim that annual textual risk scores systematically improve standalone daily volatility models. Only 20 of the 93 figure-derived diagnostics are positive, and the cross-figure mean is negative. The correct academic interpretation is therefore narrow: DAV-style textual volatility augmentation may be useful for selected firm–risk episodes, but the available peak-event evidence does not support

a general statement that DAV dominates a conventional volatility baseline. This reinforces the central design choice of the project: the main verified contribution is the cross-sectional semantic-risk topology and identity-matrix ST-GAT ablation, while DAV remains a supplemental diagnostic for frequency-mismatch and event-local behavior.

C Econometric Baseline Specification (Fama-MacBeth Regressions)

To evaluate whether the textually derived Meso-level risk vulnerabilities may carry cross-sectional return information, this study specifies a standard Fama-MacBeth (1973) two-stage regression framework. Unlike the DAV model, which analyzes isolated time-series volatility, the Fama-MacBeth methodology cross-sectionally evaluates whether exposure to specific natural-language risk factors is associated with forward returns. The final report treats this as a methodological extension because the source table for numerical Fama-MacBeth estimates is not available for audit.

C.1 Data Sources and Mathematical Formulation

The analysis utilizes the granular Meso-level risk scores extracted from the annual 10-K filings. To align with standard asset pricing literature, these annual risk exposures are held constant over the subsequent 12 months and regressed against one-month forward continuous returns ($r_{i,t+1}$), calculated using adjusted closing prices from Yahoo Finance over the 2020–2025 temporal regime. To overcome the high dimensionality and sparsity of the Meso-risk matrix (where the number of textual parameters K exceeds the cross-sectional equity universe N for specific months), the models are specified as univariate regressions.

Stage 1: Cross-Sectional Regressions

For each month t , a cross-sectional Ordinary Least Squares (OLS) regression is estimated across all active firms i :

$$r_{i,t+1} = \alpha_t + \lambda_{k,t} S_{i,k,t} + \epsilon_{i,t+1} \quad \text{for } i = 1, \dots, N_t \quad (80)$$

Where $S_{i,k,t}$ is the textual exposure of firm i to Meso-risk factor k at time t , and $\lambda_{k,t}$ is the estimated cross-sectional risk premium.

Stage 2: Time-Series Averaging

The final risk premium ($\hat{\lambda}_k$) is computed as the time-series average of the monthly estimates:

$$\hat{\lambda}_k = \frac{1}{T} \sum_{t=1}^T \hat{\lambda}_{k,t} \quad (81)$$

Standard errors are corrected using the Newey-West estimator with a 3-month lag structure, producing robust t -statistics to control for autocorrelation and heteroskedasticity.

C.2 Validation Status and Interpretation

The implemented Fama-MacBeth design is suitable as a future robustness layer for the Meso taxonomy, but the current final report does not report risk-premium coefficients, t -statistics, or p -values because the corresponding source table is absent. Until that table is available, the Fama-MacBeth module is interpreted as a validation design rather than empirical evidence for priced textual risk factors.

D Risk Contagion Network Analysis

D.1 Objective and Motivation

To validate the practical utility of the evolved taxonomy, we examine whether textual risk disclosure similarities in 10-K filings can predict risk contagion—specifically the synchronization of volatility—within the stock market. We hypothesize that firms sharing similar "risk signatures" in their disclosures (textual peers) will exhibit correlated stock price volatility.

D.2 Network Construction Methodology

We transform raw 10-K disclosures into a quantitative risk spillover model through a multi-stage pipeline:

D.2.1 Risk Exposure Vectors and Similarity

For each firm i at time t , we define a Risk Exposure Vector $\mathbf{R}_{i,t}$:

$$\mathbf{R}_{i,t} = (w_{i,t,1}, w_{i,t,2}, \dots, w_{i,t,K}) \quad (82)$$

where $w_{i,t,k}$ represents the normalized count of risk factors assigned to meso-category k . The similarity between any two firms i and j is computed using cosine similarity:

$$\text{sim}(\mathbf{R}_i, \mathbf{R}_j) = \frac{\mathbf{R}_i \cdot \mathbf{R}_j}{\|\mathbf{R}_i\| \|\mathbf{R}_j\|} \quad (83)$$

D.2.2 Spatial Weights Matrix ($\mathbf{W}$)

To define the network topology, we construct a Binary Adjacency matrix $\mathbf{A}_{i,j}$ based on a similarity threshold τ :

$$\mathbf{A}_{i,j} = \begin{cases} 1 & \text{if } \text{sim}(\mathbf{R}_i, \mathbf{R}_j) > \tau \\ 0 & \text{otherwise} \end{cases} \quad (84)$$

A value of 1 indicates a "Risk Peer" relationship exists. This is then row-normalized to create the Spatial Weights Matrix $\mathbf{W}_{i,j}$:

$$\mathbf{W}_{i,j} = \frac{\mathbf{A}_{i,j}}{\sum_{j=1}^n \mathbf{A}_{i,j}} \quad (85)$$

D.3 Spatial Autoregressive (SAR) Framework

We integrate the NLP-derived risk network with 2023 market price data using a Spatial Autoregressive (SAR) model:

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \boldsymbol{\alpha} + \boldsymbol{\epsilon} \quad (86)$$

Where:

- y : The dependent variable, represented by annualized realized volatility in the intended SAR validation design.
- W : The **Spatial Weights Matrix** derived from the evolved taxonomy.
- ρ (Rho): The **Spatial Lag coefficient**, measuring the “contagion” effect—how much a firm’s volatility is influenced by the volatility of its textual peers.
- α : The **baseline idiosyncratic volatility** of a firm, capturing the expected level of risk that remains constant regardless of the spillover effects or contagion within the textual risk network.

D.4 Validation Status and Interpretation

No auditable source table for the SAR result is available, so the report does not present SAR coefficients, p -values, or explanatory power as empirical findings. The SAR framework remains useful as a future network-validation extension: a documented rerun can test whether firms connected by textual-risk similarity also exhibit volatility co-movement after appropriate controls. In this study, the SAR appendix is therefore a methodological bridge between semantic peer construction and econometric network validation, not confirmatory evidence.

E Representative Risk Taxonomy (Sample)

Due to the high dimensionality and breadth of the evolved risk taxonomy, Table 21 presents a representative excerpt of the system’s output. This sample illustrates the hierarchical mapping between established **Macro Categories** and the more granular **Meso Categories** (Micro-level).

Table 21: Extracted Risk Taxonomy: Macro and Micro Categories

Macro Category	Micro Categories
Key Person and Succession Risk	Executive Leadership and Critical Function Dependency Risk, Executive Leadership Dependency and Transition Risk
Commodity Price & Supply Chain Volatility	Raw Material Input Cost Inflation, Human Capital & Labor Market Disruption
Strategic and Operational Market Vulnerabilities	Intellectual Property and Innovation Competition, Regulatory and Reimbursement Environment
Contractual & Regulatory Performance Risk	U.S. Government Funding & Procurement Dependency Risk, Government Contracting & Performance Risk
Data Privacy & Cybersecurity Regulatory Compliance Risk	Regulatory Compliance & Legal Liability Risk for Personal Data, Cross-Jurisdictional Data Privacy Law Compliance Risk
Human Capital Acquisition & Retention Risk	Key Personnel Dependency & Talent Competition Risk, Talent Acquisition & Retention Risk
Human Capital Management & Organizational Culture	Talent Development & Learning Systems, Diversity, Equity, Inclusion & Belonging (DEIB) Strategy & Goals
Regulatory, Legal, and Tax Compliance Volatility	Tax Law and Liability Volatility, Tax Policy and Obligation Volatility, Tax Provision and Contingency Volatility
Capital Allocation and Shareholder Returns Governance	Structural and Regulatory Constraints on Capital Deployment, Dividend Policy Discretion and Volatility Risk, Stock Price Volatility and Market Perception Risk, Corporate Governance and Controlling Shareholder Risks
Healthcare Policy, Pricing, and Reimbursement Risk	Government Program Contracting & Compliance Risk, Reimbursement and Pricing Pressure Risk
Legal, Regulatory, and Insurance Risk Exposure	Insurance Adequacy & Indemnification Risk, Litigation and Legal Contingency Risk
Strategic & Market Environment Risk	Global Regulatory & Food Safety Compliance Risk, Competitive & Market Dynamics Risk
Intellectual Property Rights & Exclusivity Management	Patent Portfolio Viability & Litigation Exposure, Third-Party IP Infringement Claims & Defense Costs, Loss of Exclusivity and Generic/Biosimilar Competition, Third-Party Intellectual Property Dependencies & Licensing, Intellectual Property Infringement & Defense
Product Commercialization and Market Dynamics	Manufacturing, Supply Chain, and Operational Capacity, Clinical Development & Safety Outcomes, Dependency on Strategic Collaborations and Market Exclusivity
Enterprise Risk Governance & Disclosure Framework	Regulatory & Disclosure Evolution Risk, Risk Factor Disclosure Framework

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Table 21 – continued from previous page

Macro Category	Micro Categories
International Trade & Geopolitical Exposure	Cross-Border Regulatory & Compliance Complexity, Global Supply Chain & Market Access Disruption
Counterparty and Credit Risk	Investment Portfolio Valuation and Impairment Risk, Credit Rating Downgrade Risk, Debt Refinancing and Liquidity Risk, Regulatory Capital & Liquidity Compliance Risk, Customer and Counterparty Credit Exposure
Human Capital & Workforce Management	Labor Cost & Workforce Stability, Talent Acquisition, Retention & Succession, Pandemic-Induced Operational & Economic Disruption, Workforce Health, Safety, & Operational Continuity
Strategic Transaction & Integration Execution Risk	Acquisition Integration & Synergy Realization Risk, Post-Transaction Integration & Synergy Realization Risk
Climate & Environmental Disruption Risk	Physical Operational Disruption from Natural Hazards, Transition & Regulatory Compliance Risk
Third-Party and Supply Chain Dependency	Manufacturing & Production Disruption, Third-Party Operational and Performance Risk, Supplier Performance and Continuity Risk
Product & Brand Integrity and Market Acceptance	Product Safety, Quality, and Public Health Perception, Consumer Preference and Health Perception Risk, Bottler Partnership and System Integrity, Competitive Positioning and Market Dynamics
Business Model and Market Evolution Risk	Product Portfolio and Technology Lifecycle Risk, Competitive Pressure and Market Positioning Risk